ANSWER KEY Reduced copy of the question page, with its answer

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$AC^2 = AD \cdot AB$

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $4/x = x/9$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If p/q = q/r, then

A. $q^2 = pr$

B. $p = r$

C. $r^2 = pq$

D. $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $q^2 = pr$

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

From $a/b = b/c$ we get $b^2 = ac$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From x/5 = 5/x we get x = 5 only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 25$, so $x = \pm 5$ ($x = 5$ for a length).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

True or false?

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

1607

QUESTION ID

S44-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

1608

QUESTION ID

S44-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Arrange four copies of the triangle inside a square of side a + b (the leftover is c², or a² + b²).

GRADING NOTE Any valid area argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

A. 5, 12, 13

B. 6, 7, 9

C. 5, 6, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

1609

QUESTION ID

S44-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5, 12, 13

GRADING NOTE 25 + 144 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

1610

QUESTION ID

S44-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Checking the theorem with three triangles proves it for all right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

1611

QUESTION ID

S44-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples are not proofs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with altitude CD to hypotenuse AB, AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

1613

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C and altitude CD, prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

1614

QUESTION ID

S44-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$CD^2 = AD \cdot DB$

GRADING NOTE REQUIRED WORK: Show △ACD ~ △CBD and write a proportion.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △CDB (AA: right angles at D, and ∠A ≅ ∠BCD since both are complementary to ∠B); AD/CD = CD/DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

True or false?

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

1615

QUESTION ID

S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. AD = 2 and AB = 8. Find AC.

A. 4

B. $2\sqrt{2}$

C. 6

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

1616

QUESTION ID

S44-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $AC^2 = AD \cdot AB = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

1617

QUESTION ID

S44-F1-Q5

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. 144 = 9x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab\cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

1619

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2$ (the Pythagorean Theorem), since $\cos 90° = 0$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

1620

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 25 + 64 − 40 = 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

True or false?

If $a^2 + b^2 > c^2$, the angle opposite c is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

1621

QUESTION ID
S44-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is acute; the angle is obtuse when $a^2 + b^2 < c^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

A triangle has sides 4, 5, 7. The largest angle is

A. right

B. impossible to tell

C. obtuse

D. acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

1622

QUESTION ID

S44-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE $16 + 25 = 41 < 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

1623

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Right

(49 + 576 = 625)

GRADING NOTE Form required: the classification with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

Use similarity criteria to solve problems in geometric figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Similarity criteria: AA, SAS, SSS	**HSG-SRT.A.3**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Identify corresponding sides in overlapping triangles	**HSG-SRT.A.2**
FORWARD 1	Solve multistep problems with two or more pairs of similar triangl...	**HSG-SRT.B.5**
FORWARD 2	Solve similarity problems with algebra (unknowns on both sides)	**HSG-SRT.B.5**

SET 45 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

1626

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

BC = 20; △ADE ~ △ABC by AA.

GRADING NOTE Form required: the length and the criterion. 6/15 = 8/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

1627

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 m; AA (the shared angle at the tip of the shadow, and two right angles).

GRADING NOTE Form required: the height and the criterion. The tip of the shadow is 5 m from the post: 1.6/2 = h/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: △PQR with PQ = 8, QR = 12, ∠Q = 50°; △STU with ST = 6, TU = 9, ∠T = 50°. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

1628

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes, SAS similarity (8/6 = 12/9 = 4/3); perimeter ratio 4 : 3.

GRADING NOTE Form required: the criterion and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

A surveyor finds a river's width w with similar right triangles: legs 12 m and 18 m on the small triangle, 12 m corresponding to 40 m on the large one. What is w (the side matching 18 m)?

A. 26.7 m

B. 46 m

C. 54 m

D. 60 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

1629

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 60 m

GRADING NOTE $18 \cdot 40/12 = 60$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

True or false?

If two triangles have sides 3, 5, 7 and 6, 10, 15, they are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

1630

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $15/7 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

1632

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

1633

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

SSS similarity

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

A. Two pairs of congruent angles

B. One pair of congruent angles

C. One pair of proportional sides

D. Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

1634

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Two pairs of congruent angles

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

True or false?

SAS similarity requires the congruent angle to be between the proportional sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

1635

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

1636

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need another congruent angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

1638

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

1639

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 12/40 = 18/x.

A. 26.7

B. 54

C. 46

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

1640

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60

GRADING NOTE x = 18 · 40/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3/4 = 9/12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

1641

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/8 = 10/14

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

1642

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/16 would be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

△ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

1644

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

True or false?

In △ADE ~ △ABC, side AD corresponds to side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

1645

QUESTION ID

S45-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. In △ADE ~ △ABC, the side of the large triangle that corresponds to AD is

A. DB

B. AC

C. BC

D. AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

1646

QUESTION ID
S45-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. AB

GRADING NOTE Whole side, not the piece DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

AD = 6, DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

1647

QUESTION ID

S45-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B3-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In △ADE ~ △ABC, DB corresponds to DE.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

1648

QUESTION ID

S45-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DB is not a side of △ADE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

1650

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

AE = 12, EC = 8

GRADING NOTE Form required: both. AE/EC = 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

1651

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.25 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

True or false?

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

1652

QUESTION ID
S45-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Shared sides link them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with AD = 2, DB = 6; and FG ∥ BC farther down with AF = 5. If BC = 16, find FG.

A. 10

B. 12

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

1653

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE FG/BC = AF/AB = 5/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

1654

QUESTION ID

S45-F1-Q5

TYPE

Short answer

CORRECT ANSWER

9 m

GRADING NOTE Any equivalent form. 1.5/2 = h/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with AD = x, DB = 6, AE = 4, EC = x + 2. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

1656

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. x(x + 2) = 24, x > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Similar triangles have corresponding sides x + 1 and 2x − 3 with scale factor 1.5 (larger to smaller is 2x − 3 to x + 1). Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

1657

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. 2x − 3 = 1.5(x + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

True or false?

A proportion with the unknown in two places can lead to a quadratic equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

1658

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiplying gives products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

In right triangle ABC with the right angle at C, the altitude CD meets hypotenuse AB at D. AD = x, DB = x + 5, and CD = 6. Find x.

A. 4

B. 3

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

1659

QUESTION ID

S45-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE x(x + 5) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

1660

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4/x = x/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

Use congruence criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Choose and apply SSS, SAS, ASA, AAS	**HSG-CO.B.8**
BACKWARD 2	Find hidden parts: shared sides, vertical angles, right angles	**HSG-CO.C.9**
BACKWARD 3	Use corresponding parts to finish a proof	**HSG-CO.B.7**
FORWARD 1	Prove relationships that need two congruences in a row	**HSG-SRT.B.5**
FORWARD 2	Decide which given data are sufficient for a proof, and supply a m…	**HSG-SRT.B.5**

SET 46 • HSG-SRT.B.5

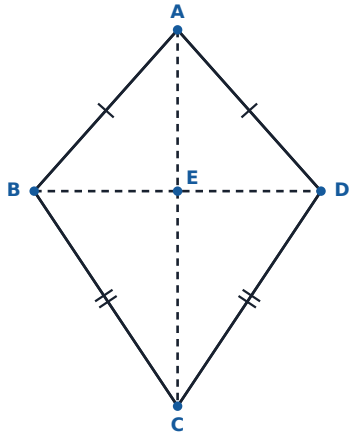
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

1663

QUESTION ID

S46-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC ⊥ BD

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △ADC (SSS) to get ∠BAE ≅ ∠DAE; then △ABE ≅ △ADE (SAS); then the angles at E form a congruent linear pair.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = AD, CB = CD, AC shared); ∠BAC ≅ ∠DAC; SAS on △ABE and △ADE (AE shared); ∠AEB ≅ ∠AED, a linear pair, so 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

1664

QUESTION ID

S46-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AC ∥ DB

GRADING NOTE REQUIRED WORK: Prove △AMC ≅ △BMD (SAS with vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM, CM = DM (bisect), ∠AMC ≅ ∠BMD (vertical); SAS; ∠CAM ≅ ∠DBM; these are alternate interior angles for AC and DB with transversal AB; converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

1665

QUESTION ID

S46-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

PD = PE

GRADING NOTE

REQUIRED WORK: Prove △PBD ≅ △PBE by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PBD ≅ ∠PBE (bisector), ∠PDB ≅ ∠PEB (right angles), PB shared; AAS; PD = PE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. M is the midpoint of BC. Prove that AM bisects ∠BAC and AM ⊥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

1666

QUESTION ID

S46-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

∠BAM ≅ ∠CAM and AM ⊥ BC

GRADING NOTE REQUIRED WORK: Prove △ABM ≅ △ACM (SSS), then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: SSS; ∠BAM ≅ ∠CAM (bisects); ∠AMB ≅ ∠AMC form a linear pair, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

To prove that two segments in a figure are congruent, the most common first step is to

A. prove two triangles containing them are congruent

B. measure them carefully

C. show the figure is symmetric by looking

D. assume they are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

1667

QUESTION ID
S46-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. prove two triangles containing them are congruent

GRADING NOTE Then use corresponding parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

1669

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

1670

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

AAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

A. A pair of congruent sides and the congruent included angles

B. Two pairs of congruent sides not including the angle

C. Nothing more is needed

D. Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

1671

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A pair of congruent sides and the congruent included angles

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A shared side is a valid pair for any criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

1672

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

True or false?

AAA proves congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

1673

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

1675

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠BMD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

True or false?

Perpendicular lines give congruent right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

1676

QUESTION ID

S46-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

A. All angles are equal

B. The shared side is congruent to itself

C. The triangles are right

D. Nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

1677

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The shared side is congruent to itself

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

1678

QUESTION ID

S46-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠ABP and ∠PBC (for bisector ray BP)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Vertical angles are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

1679

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△ABC ≅ △ADC. Which angle corresponds to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

1681

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠DAC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

True or false?

Once triangles are congruent, all corresponding parts are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

1682

QUESTION ID

S46-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△PBD ≅ △PBE. Which conclusion follows?

A. PD = PE

B. PB = PD

C. BD = PE

D. ∠D ≅ ∠B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

1683

QUESTION ID

S46-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. PD = PE

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

1684

QUESTION ID

S46-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

True or false?

Corresponding parts can be used before the triangles are proved congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

1685

QUESTION ID

S46-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only after.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

1687

QUESTION ID

S46-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AE = CE

GRADING NOTE

REQUIRED WORK: Prove △ABC ≅ △CDA (SSS) to get parallel sides, then △ABE ≅ △CDE (ASA).

Any valid proof (paragraph, two-column, or flow). Essential: SSS gives alternate interior angles congruent (AB ∥ DC); then ASA (or AAS) for △ABE and △CDE gives AE = CE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. D is on AB and E is on AC with BD = CE. Prove that BE = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

1688

QUESTION ID

S46-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

BE = CD

GRADING NOTE REQUIRED WORK: Show AD = AE, then prove △ABE ≅ △ACD by SAS (shared ∠A).

Any valid proof (paragraph, two-column, or flow). Essential: AD = AB − BD = AC − CE = AE; ∠A shared; AB = AC; SAS; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

True or false?

Some proofs need one congruence to find a part used in a second congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

1689

QUESTION ID
S46-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Chained congruences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

A. measure

B. end the proof

C. supply a pair of parts for the second congruence

D. restart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

1690

QUESTION ID
S46-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. supply a pair of parts for the second congruence

GRADING NOTE Chaining.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

1691

QUESTION ID
S46-F1-Q5

TYPE
Short answer

CORRECT ANSWER
That would assume what must be proved (circular reasoning).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △PSR (sharing PR), PQ = PS. What one more fact would prove ∠QPR ≅ ∠SPR? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

1693

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

QR = SR (then SSS gives the congruence and the angles correspond).

GRADING NOTE Any sufficient condition with explanation (∠PRQ ≅ ∠PRS would give SSA, which is not enough).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

1694

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

False in general: only the diagonal BD is bisected (by AC); AC is bisected only if the kite is a rhombus. Example: A(0, 3), B(−1, 0), C(0, −1), D(1, 0).

GRADING NOTE Any valid counterexample or explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

True or false?

Given AB = DE, BC = EF, and ∠C ≅ ∠F, the triangles must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

1695

QUESTION ID
S46-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE SSA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given ∠A ≅ ∠D and AB = DE in △ABC and △DEF, which extra fact gives ASA?

A. ∠C ≅ ∠F

B. BC = EF

C. AC = DF

D. ∠B ≅ ∠E

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

1696

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠B ≅ ∠E

GRADING NOTE The angle at the other end of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

1697

QUESTION ID
S46-F2-Q5

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠D (the included angle)

GRADING NOTE Form required: the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

Use similarity criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Prove triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Turn a proportion into a product	**7.RP.A.2**
BACKWARD 3	Match corresponding sides of similar triangles	**HSG-SRT.A.2**
FORWARD 1	Use proved product relationships to compute	**HSG-SRT.B.5**
FORWARD 2	Prove relationships in circles with similar triangles	**HSG-C.A.2**

SET 47 • HSG-SRT.B.5

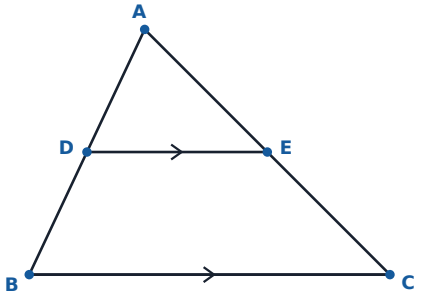
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with D on AB and E on AC. Prove that AD · AC = AE · AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

1700

QUESTION ID

S47-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD · AC = AE · AB

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: AA (shared ∠A; corresponding angles); AD/AB = AE/AC; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

1701

QUESTION ID

S47-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE REQUIRED WORK: Prove △PAD ~ △PCB (AA: the given angles and vertical angles), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PAD ≅ ∠PCB (given), ∠APD ≅ ∠CPB (vertical); AA; PA/PC = PD/PB; cross-multiply.

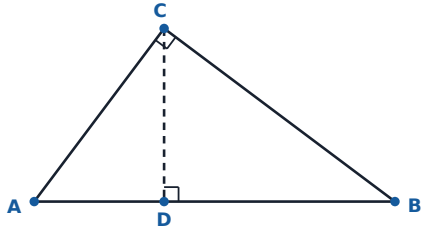
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

1702

QUESTION ID

S47-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AC · BC = AB · CD

GRADING NOTE REQUIRED WORK: Prove △ACB ~ △ADC (or △CDB) and use the ratio of corresponding sides.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (AA); DC/CB = AC/AB; cross-multiply. Alternative: both products equal twice the area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

1703

QUESTION ID

S47-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AE/EC = AB/DC

GRADING NOTE REQUIRED WORK: Prove △ABE ~ △CDE by AA (alternate interior angles or vertical angles).

Any valid proof (paragraph, two-column, or flow). Essential: ∠EAB ≅ ∠ECD and ∠EBA ≅ ∠EDC (alternate interior, AB ∥ DC); AA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

To prove a product relationship such as PA · PB = PC · PD, which is the usual strategy?

A. Use the Pythagorean Theorem only

B. Measure the four lengths

C. Find two similar triangles whose sides appear in the products

D. Assume the product is true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

1704

QUESTION ID

S47-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Find two similar triangles whose sides appear in the products

GRADING NOTE A proportion from similar triangles becomes a product by cross-multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Vertical angles can be one pair of congruent angles in an AA proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

1706

QUESTION ID

S47-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

1707

QUESTION ID
S47-B1-Q2

TYPE
Short answer

CORRECT ANSWER
AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallel lines in a figure usually supply

A. equal side lengths

B. a scale factor

C. right angles

D. congruent corresponding or alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

1708

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. congruent corresponding or alternate interior angles

GRADING NOTE Angle pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

1709

QUESTION ID

S47-B1-Q4

TYPE

Short answer

CORRECT ANSWER

PC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

True or false?

Similarity requires all three pairs of sides to be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

1710

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

1712

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

ad = bc

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

1713

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If x/3 = 4/6, then

A. 6x = 12

B. x = 8

C. 3x = 24

D. 4x = 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

1714

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6x = 12

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

a/b = c/d implies ab = cd.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

1715

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It implies ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

True or false?

2/5 = 6/15 implies 2 · 15 = 5 · 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

1716

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 30 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

△ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

1718

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

AD/AB = AE/AC

GRADING NOTE Any equivalent proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

True or false?

In △ABE ~ △CDE, AB corresponds to CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

1719

QUESTION ID

S47-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ACB ~ △ADC. Which side corresponds to CB?

A. DC

B. AD

C. AB

D. AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

1720

QUESTION ID

S47-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. DC

GRADING NOTE A ↔ A, C ↔ D, B ↔ C, so CB ↔ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B3-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

1721

QUESTION ID

S47-B3-Q4

TYPE

Short answer

CORRECT ANSWER

XY/UV = YZ/VW = XZ/UW

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

True or false?

Any two sides of similar triangles are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

1722

QUESTION ID
S47-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

1724

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4 · 9 = 6 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

1725

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Any equivalent form. 6 · 8 = 10 · CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

True or false?

If two chords meet inside a circle, the products of their segments are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

1726

QUESTION ID
S47-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Proved by similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

A. 8

B. 4

C. 13.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

1727

QUESTION ID

S47-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE AE/EC = 18/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

DE ∥ BC in △ABC with AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

1728

QUESTION ID

S47-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. AD · AC = AE · AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given ∠PTA ≅ ∠PBT, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

1730

QUESTION ID

S47-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$PT^2 = PA \cdot PB$

GRADING NOTE REQUIRED WORK: Prove △PTA ~ △PBT (AA), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and the given angles; AA; PT/PB = PA/PT; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

1731

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. 64 = 4 · PB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

True or false?

Two secants from an outside point P satisfy PA + PB = PC + PD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

1732

QUESTION ID

S47-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The products are equal: PA · PB = PC · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

A. 7

B. 1

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

1733

QUESTION ID

S47-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE 36 = 4 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

1734

QUESTION ID
S47-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Angles that intercept the same arc are congruent (inscribed angle theorem), which supplies the AA pairs.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

Understand trigonometric ratios as properties of an angle, by similarity

MAIN	Geometry main question	**HSG-SRT.C.6**
BACKWARD 1	Similar triangles have equal side ratios within each triangle	**HSG-SRT.A.2**
BACKWARD 2	Name the opposite side, adjacent side, and hypotenuse for an ang	**HSG-SRT.A.3**
BACKWARD 3	Write and simplify ratios	**6.RP.A.1**
FORWARD 1	Find trig ratios from one ratio, using a similar right triangle	**HSG-SRT.C.6**
FORWARD 2	The unit circle extends the ratios beyond acute angles	**HSF-TF.A.2**

SET 48 • HSG-SRT.C.6

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Right triangles ABC and DEF have right angles at C and F and ∠A ≅ ∠D. Explain why BC/AB = EF/DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

1737

QUESTION ID

S48-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

BC/AB = EF/DE

GRADING NOTE

REQUIRED WORK: Use AA similarity, then the proportion of corresponding sides, rearranged.

Any valid proof (paragraph, two-column, or flow). Essential: right angles and ∠A ≅ ∠D give AA similarity; BC/EF = AB/DE; rearranging gives BC/AB = EF/DE. So the ratio depends only on the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

1738

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

7.5; the triangles are similar (AA), so opposite/hypotenuse = 4/8 = ½ in both.

GRADING NOTE Form required: the length and the similarity reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

1739

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin P = 12/13, cos P = 5/13, tan P = 12/5

GRADING NOTE Form required: all three ratios (any equivalent fractions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

Why is sin 40° the same in every right triangle with a 40° angle?

A. All such triangles are similar, so the opposite-to-hypotenuse ratio is the same

B. Every hypotenuse is 1

C. The sides are always 3, 4, 5

D. Sine depends on the triangle's size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

1740

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. All such triangles are similar, so the opposite-to-hypotenuse ratio is the same

GRADING NOTE Similarity fixes the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

True or false?

Doubling every side of a right triangle doubles the tangent of its acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

1741

QUESTION ID
S48-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ratio stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar.
Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

1743

QUESTION ID
S48-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Both equal 0.6.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

True or false?

If △ABC ~ △DEF, then AB/AC = DE/DF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

1744

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Ratios within each triangle match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △DEF with BC/AB = 0.8. Find EF/DE.

A. Cannot tell

B. 0.8

C. 1.25

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

1745

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.8

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

1746

QUESTION ID

S48-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

True or false?

Similar triangles have equal side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

1747

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

1749

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

1750

QUESTION ID
S48-B2-Q2

TYPE
Short answer

CORRECT ANSWER
AC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

1750

ANSWER KEY Reduced copy of the question page, with its answer

The hypotenuse is always

A. opposite the right angle

B. next to the right angle

C. the shortest side

D. opposite the smallest angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

1751

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. opposite the right angle

GRADING NOTE Longest side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

The side opposite ∠A is adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

1752

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE BC is next to ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

The opposite side of an angle depends on the triangle's size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

1753

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It depends on position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

1755

QUESTION ID

S48-B3-Q1

TYPE

Short answer

CORRECT ANSWER

12/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

1756

QUESTION ID

S48-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

Which equals 0.75?

A. 7/5

B. 3/7

C. 3/4

D. 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

1757

QUESTION ID

S48-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

GRADING NOTE 3 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/13 ≈ 0.385

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

1758

QUESTION ID

S48-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

True or false?

A ratio changes when both quantities are tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

1759

QUESTION ID

S48-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

For an acute angle θ, sin θ = 3/5. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

1761

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

cos θ = 4/5, tan θ = 3/4

GRADING NOTE Form required: both. A 3-4-5 triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 5/12 for an acute angle θ. Find sin θ.

sin θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

1762

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

1763

QUESTION ID

S48-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 0.6. Which is sin θ (θ acute)?

A. 0.8

B. 1.6

C. 0.4

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

1764

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.8

GRADING NOTE √(1 − 0.36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\sin\theta < 1$ for every acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

1765

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The opposite leg is shorter than the hypotenuse.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

On a unit circle, the point for angle θ is (cos θ, sin θ).
Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

1767

QUESTION ID
S48-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

1768

QUESTION ID
S48-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(√3/2, ½)

GRADING NOTE Form required: exact values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

True or false?

For acute angles, the unit-circle definition of sine gives a different value from the right-triangle definition.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

1769

QUESTION ID
S48-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They agree: the hypotenuse is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 150° on the unit circle equals

A. −½

B. −√3/2

C. √3/2

D. ½

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

1770

QUESTION ID
S48-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. ½

GRADING NOTE Point (−√3/2, ½).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

1771

QUESTION ID

S48-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The point on the unit circle at 120° has a y-coordinate, which defines sin 120°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

Explain and use the relationship between the sine and cosine of complementary angles

MAIN	Geometry main question	**HSG-SRT.C.7**
BACKWARD 1	Complementary angles add to 90°	**7.G.B.5**
BACKWARD 2	Opposite and adjacent sides depend on the chosen angle	**HSG-SRT.C.6**
BACKWARD 3	Angle sum in a right triangle	**8.G.A.5**
FORWARD 1	Relationships between tangents of complementary angles	**HSG-SRT.C.7**
FORWARD 2	The Pythagorean identity	**HSF-TF.C.8**

SET 49 • HSG-SRT.C.7

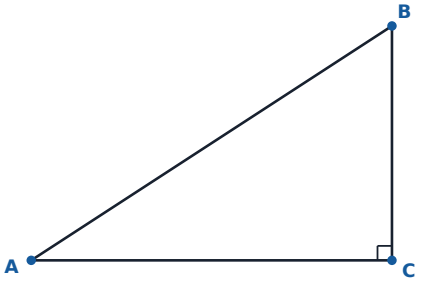
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space



Conclusion: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

1774

QUESTION ID

S49-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

sin A = BC/AB = cos B, and A and B are complementary.

GRADING NOTE REQUIRED WORK: Name the side opposite ∠A and the side adjacent to ∠B, and use the angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = BC/AB (opposite over hypotenuse); cos B = BC/AB (BC is adjacent to ∠B); so they are equal; ∠A + ∠B = 90°, so sin A = cos(90° − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 25° = cos x°, where 0 < x < 90. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

1775

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

65

GRADING NOTE Any equivalent form. Complementary angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

θ is acute and cos θ = 0.6. Find sin(90° − θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

1776

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

Both angles are acute and sin(2x + 10)° = cos(3x)°.
Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

1777

QUESTION ID
S49-M-Q4

TYPE
Short answer

CORRECT ANSWER
16

GRADING NOTE Any equivalent form. 2x + 10 + 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression equals cos 72°?

A. cos 18°

B. sin 72°

C. sin 18°

D. tan 18°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

1778

QUESTION ID
S49-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. sin 18°

GRADING NOTE 72 + 18 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

1780

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

53°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

True or false?

The two acute angles of a right triangle are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

1781

QUESTION ID

S49-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 2x are complementary. x =

A. 90°

B. 45°

C. 60°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

1782

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30°

GRADING NOTE 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The complement of θ is 90° − θ. If θ = 64°, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

1783

QUESTION ID

S49-B1-Q4

TYPE

Short answer

CORRECT ANSWER

26°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Complementary angles must be adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

1784

GRADING NOTE They need only add to 90°.

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

1786

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

True or false?

The side opposite ∠A is the side adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

1787

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

cos B equals

A. AC/AB

B. BC/AB

C. AC/BC

D. BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

1788

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. BC/AB

GRADING NOTE Adjacent over hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

1789

QUESTION ID
S49-B2-Q4

TYPE
Short answer

CORRECT ANSWER
BC/AB

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin A and cos A are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

1790

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when A = 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

1792

QUESTION ID

S49-B3-Q1

TYPE

Short answer

CORRECT ANSWER

62°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

1793

QUESTION ID

S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, the acute angles are x and x + 20. x =

A. 70°

B. 45°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

1794

QUESTION ID

S49-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 2x + 20 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

1795

QUESTION ID

S49-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The angles add to 180° and the right angle uses 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

True or false?

The two acute angles of an isosceles right triangle are 45° each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

1796

QUESTION ID

S49-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal and complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

1798

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

tan θ = opposite/adjacent and tan(90° − θ) = adjacent/opposite, which are reciprocals.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan 20° ≈ 0.364. Estimate tan 70°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

1799

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 2.75

(1 ÷ 0.364)

GRADING NOTE Any value 2.74-2.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin \theta = \cos \theta$ exactly when $\theta = 45°$ (for acute θ).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

1800

QUESTION ID

S49-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then $90° - \theta = \theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals 1 for every acute θ?

A. $\tan\theta + \tan(90° - \theta)$

B. $\tan\theta \cdot \tan(90° - \theta)$

C. $\sin\theta \cdot \cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

1801

QUESTION ID
S49-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $\tan\theta \cdot \tan(90° - \theta)$

GRADING NOTE Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that sin θ / cos(90° − θ) = 1 for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

1802

QUESTION ID

S49-F1-Q5

TYPE

Short answer

CORRECT ANSWER

cos(90° − θ) = sin θ, so the quotient is sin θ / sin θ = 1.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle θ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

1804

QUESTION ID

S49-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(\sin \theta)^2 + (\cos \theta)^2 = 1$

GRADING NOTE REQUIRED WORK: Use a right triangle with legs a, b and hypotenuse c and the Pythagorean Theorem.

Any valid proof (paragraph, two-column, or flow). Essential: $\sin \theta = a/c$, $\cos \theta = b/c$; $a^2 + b^2 = c^2$; divide by c^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.28 for acute θ. Find cos θ.

cos θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

1805

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.96

GRADING NOTE Any equivalent form. √(1 − 0.0784).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 35° + \cos 35° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

1806

QUESTION ID
S49-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The squares add to 1; $\sin 35° + \cos 35° \approx 1.39$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 12/13 (θ acute). sin θ =

A. 1/13

B. 5/13

C. 12/5

D. 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

1807

QUESTION ID

S49-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/13

GRADING NOTE √(1 − 144/169).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

1808

QUESTION ID
S49-F2-Q5

TYPE
Short answer

CORRECT ANSWER
sin 70° = cos 20°, so the sum is $(\sin 20°)^2 + (\cos 20°)^2 = 1$.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown sides of right triangles with trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Choose the trigonometric ratio that links the known and unknown.	**HSG-SRT.C.6**
BACKWARD 2	Solve one-variable equations with a multiplication or division	**HSA-REI.B.3**
BACKWARD 3	Round to a stated precision and keep track of units	**HSN-Q.A.3**
FORWARD 1	Find a height from two angle-of-elevation observations	**HSG-SRT.C.8**
FORWARD 2	Find a feasible location under a clearance constraint	**HSG-MG.A.3**

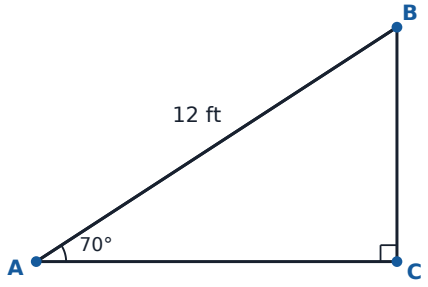
SET 50 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?



Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

1811

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

11.3 ft

GRADING NOTE Rounding to 0.1 required. 12 sin 70° = 11.2763.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

1812

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

18.0 ft

GRADING NOTE Rounding to 0.1 required. 2.5 ÷ sin 8° = 17.9632.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

1813

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Rounding to 0.1 required. 25 tan 38° = 19.5321.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

A. 78 m

B. 46 m

C. 39 m

D. 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

1814

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 46 m

GRADING NOTE Opposite = hypotenuse × sin 50° = 45.96.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

1815

QUESTION ID
S50-M-Q5

TYPE
Short answer

CORRECT ANSWER
17.0

GRADING NOTE Rounding to 0.1 required. 20 cos 32° = 16.9610.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the hypotenuse and an angle and want the opposite side. Which ratio?

A. none

B. sine

C. cosine

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

1817

QUESTION ID

S50-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. sine

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

You know the adjacent side and an angle and want the opposite side. Which ratio?

A. none

B. tangent

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

1818

QUESTION ID

S50-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

1819

QUESTION ID

S50-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

True or false?

To find the hypotenuse from the opposite side and the angle, use sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

1820

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE sin A = opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent uses the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

1821

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent uses the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/12 = 0.94.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

1823

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

11.28

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.14 = 2.5/x.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

1824

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

17.86

GRADING NOTE Any value 17.8-17.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin 30° = h/10, then h =

A. 10 sin 30°

B. 10 ÷ sin 30°

C. sin 30° ÷ 10

D. 30 ÷ 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

1825

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 sin 30°

GRADING NOTE Multiply both sides by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From 0.5 = 4/x we get x = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

1826

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE x = 4 ÷ 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

True or false?

From tan 40° = 20/x we get x = 20 tan 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

1827

QUESTION ID
S50-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE x = 20 ÷ tan 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

1829

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

11.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

1830

QUESTION ID

S50-B3-Q2

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

Rounding in the middle of a calculation can change the final answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

1831

QUESTION ID

S50-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Round at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A height is measured in feet and the angle in degrees. The answer's unit is

A. degrees

B. feet per degree

C. feet

D. no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

1832

QUESTION ID
S50-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. feet

GRADING NOTE Lengths keep their units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

True or false?

17.96 rounded to the nearest tenth is 17.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

1833

QUESTION ID

S50-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 18.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

1835

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

42.0 m

GRADING NOTE Rounding to 0.1 required. h = 40 ÷ (1/tan 25° − 1/tan 40°) = 41.984. Two equations: h = x tan 40° and h = (x + 40) tan 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. How far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

1836

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50.0 m

GRADING NOTE Rounding to 0.1 required. x = h ÷ tan 40° = 50.034.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

True or false?

With only one angle of elevation and no distance, the height cannot be found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

1837

QUESTION ID
S50-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE One equation has two unknowns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

A. h = 40 tan 25°

B. h = x tan 25° and h = (x + 40) tan 40°

C. h = x tan 40° and h = (x + 40) tan 25°

D. h = x sin 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

1838

QUESTION ID

S50-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. h = x tan 40° and h = (x + 40) tan 25°

GRADING NOTE The closer point has the larger angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

1839

QUESTION ID

S50-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The same height over a shorter horizontal distance gives a larger tangent, so a larger angle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest allowed horizontal run, to the nearest foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

1841

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

36 ft

GRADING NOTE Form required: a whole number of feet that meets the constraint (round up). 3 ÷ tan 4.8° = 35.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. How far from the pole can that point be, at most, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

1842

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.5 m

GRADING NOTE Rounding to 0.1 required. 6 ÷ tan 20° = 16.485; farther points need smaller angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

True or false?

Rounding UP the run of a ramp with a maximum angle can make the ramp too steep.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

1843

QUESTION ID

S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A longer run makes the ramp less steep; rounding the run down is the risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. The shortest ladder that works is about

A. 10.0 ft

B. 9.7 ft

C. 2.7 ft

D. 10.4 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

1844

QUESTION ID

S50-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10.4 ft

GRADING NOTE At exactly 75°, length = 10 ÷ sin 75° = 10.35; a smaller angle needs a longer ladder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, to the nearest tenth of a foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

1845

QUESTION ID

S50-F2-Q5

TYPE

Short answer

CORRECT ANSWER

11.9 ft

GRADING NOTE Rounding to 0.1 required. 9 ÷ tan 37° = 11.943.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown angles with inverse trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Inverse trig undoes a trig ratio	**HSG-SRT.C.6**
BACKWARD 2	Choose the ratio from the two known sides	**HSG-SRT.C.6**
BACKWARD 3	Use degree mode and round angles	**HSN-Q.A.3**
FORWARD 1	Solve a right triangle completely from two sides	**HSG-SRT.C.8**
FORWARD 2	Find directions (bearings) with inverse trigonometry	**HSG-SRT.C.8**

SET 51 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

1848

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.8°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(1/12) = 4.764°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

1849

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

66.4°

GRADING NOTE Rounding to 0.1° required. $\sin^{-1}(5.5/6) = 66.444°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

1850

QUESTION ID
S51-M-Q3

TYPE
Short answer

CORRECT ANSWER
35.5°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(10/14) = 35.538°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

A. 73.7°

B. 16.3°

C. 32.5°

D. 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

1851

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16.3°

GRADING NOTE $\tan^{-1}(7/24) = 16.260°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

1852

QUESTION ID

S51-M-Q5

TYPE

Short answer

CORRECT ANSWER

3.4°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(0.06) = 3.434°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.5 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

1854

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 1 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

1855

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which gives an angle from a ratio?

A. cos

B. tan

C. sin

D. $\sin^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

1856

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sin^{-1}$

GRADING NOTE Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0.5) = 60°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

1857

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE cos 60° = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

1858

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the angle whose sine is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

A. none

B. $\cos^{-1}$

C. $\sin^{-1}$

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

1860

QUESTION ID
S51-B2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. $\sin^{-1}$

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You know both legs. Which inverse function finds an acute angle?

A. $\sin^{-1}$

B. $\tan^{-1}$

C. $\cos^{-1}$

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

1861

QUESTION ID
S51-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. $\tan^{-1}$

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

1862

QUESTION ID

S51-B2-Q3

TYPE

Short answer

CORRECT ANSWER

tan θ = 5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

True or false?

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

1863

QUESTION ID
S51-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

1864

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin^{-1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

1866

QUESTION ID
S51-B3-Q1

TYPE
Short answer

CORRECT ANSWER
4.8°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

1867

QUESTION ID
S51-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An acute angle of a right triangle must be

A. more than 90°

B. any angle

C. exactly 45°

D. between 0° and 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

1868

QUESTION ID

S51-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. between 0° and 90°

GRADING NOTE Acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

1869

QUESTION ID

S51-B3-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer; check the calculation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

True or false?

35.537° to the nearest tenth is 35.6°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

1870

QUESTION ID
S51-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 35.5°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

1872

QUESTION ID
S51-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Hypotenuse 16.6; angles 32.7° and 57.3°

GRADING NOTE Rounding to 0.1 required for all three; the angles add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

1873

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Other leg 15; angles 28.1° and 61.9°

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

True or false?

Once one acute angle is known, the other is 90° minus it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

1874

QUESTION ID

S51-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

A. 45° and 45°

B. 50° and 40°

C. 30° and 60°

D. 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

1875

QUESTION ID

S51-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 45° and 45°

GRADING NOTE Isosceles right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

1876

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The acute angles add to 90°, $a^2 + b^2 = c^2$, and each ratio matches its angle.

GRADING NOTE Any reasonable checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

1878

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

32.0°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

1879

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

53.1°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

True or false?

Bearing problems become right-triangle problems when the motions are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

1880

QUESTION ID

S51-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE East-west and north-south legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

A. 22.6°

B. 24.6°

C. 67.4°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

1881

QUESTION ID

S51-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 22.6°

GRADING NOTE $\tan^{-1}(50/120)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A ship is 10 km east and 10 km south of port. Find the straight-line distance and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

1882

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14.1 km; 45°

GRADING NOTE Form required: both. $10\sqrt{2} \approx 14.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Use the Pythagorean Theorem to find a side	**8.G.B.7**
BACKWARD 2	Interpret angles of elevation and depression	**HSG-SRT.C.6**
BACKWARD 3	Choose sine, cosine, or tangent	**HSG-SRT.C.6**
FORWARD 1	Solve problems that need two right triangles	**HSG-SRT.C.8**
FORWARD 2	Solve non-right triangles with the Law of Sines	**HSG-SRT.D.11**

SET 52 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

1885

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

171.6 m from the base; 189.3 m from the observer

GRADING NOTE Rounding to 0.1 required for both. Horizontal = 80 ÷ tan 25°; line of sight = 80 ÷ sin 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

1886

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

8022 m horizontal; 8267 m line of sight

GRADING NOTE Rounding to the nearest meter required for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

1887

QUESTION ID
S52-M-Q3

TYPE
Short answer

CORRECT ANSWER
Run 14.9 ft; angle 7.7°

GRADING NOTE Rounding required as stated. Run = $\sqrt{(15^2 - 2^2)}$; angle = $\sin^{-1}(2/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

A. 26.6 ft; 68.9°

B. 16.0 ft; 21.1°

C. 23.3 ft; 21.1°

D. 23.3 ft; 68.9°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

1888

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 23.3 ft; 68.9°

GRADING NOTE Height $\sqrt{(25^2 - 9^2)}$; angle $\cos^{-1}(9/25)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

1889

QUESTION ID
S52-M-Q5

TYPE
Short answer

CORRECT ANSWER
Legs 16.1 and 19.2; other angle 50°

GRADING NOTE Form required: both legs and the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

1891

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

23.3

GRADING NOTE Any equivalent form. √(625 − 81).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

1892

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

17

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, the hypotenuse is the longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

1893

QUESTION ID

S52-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is opposite the largest angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

Hypotenuse 13, leg 5. Other leg?

A. √194

B. 12

C. 8

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

1894

QUESTION ID

S52-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ works for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

1895

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

True or false?

An angle of depression is measured down from the horizontal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

1897

QUESTION ID

S52-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

1898

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25°

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B2-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

Which side is horizontal in an angle-of-elevation triangle?

A. The hypotenuse

B. The leg along the ground

C. The vertical leg

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

1899

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The leg along the ground

GRADING NOTE Level ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

1900

QUESTION ID

S52-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Elevation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angle of depression is measured from the vertical line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

1901

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE From the horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

A. none

B. tangent

C. sine

D. cosine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

1903

QUESTION ID

S52-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. sine

GRADING NOTE sin = opp/hyp.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

Known: adjacent side and angle. Wanted: opposite side. Ratio?

A. cosine

B. tangent

C. none

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

1904

QUESTION ID

S52-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE tan = opp/adj.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

1905

QUESTION ID

S52-B3-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

True or false?

Two different ratios can lead to the same correct side if used correctly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

1906

QUESTION ID

S52-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, finding the third side by trig or Pythagoras.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent relates the hypotenuse and a leg.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

1907

QUESTION ID

S52-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent relates the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

1909

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

33.8 m

GRADING NOTE Rounding to 0.1 required. Street width = 20 ÷ tan 40° = 23.835; height = 20 + width · tan 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

1910

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

44.1 m

GRADING NOTE Rounding to 0.1 required. h/tan 35° + h/tan 50° = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

True or false?

A shared side links two right triangles in such problems.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

1911

QUESTION ID

S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, a common height or distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

Two observers 100 m apart on opposite sides of a tower of height h see its top at angles of elevation 35° and 50°. Which equation is correct?

A. h = 100 tan 85°

B. h/sin 35° + h/sin 50° = 100

C. h tan 35° + h tan 50° = 100

D. h/tan 35° + h/tan 50° = 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

1912

QUESTION ID

S52-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. h/tan 35° + h/tan 50° = 100

GRADING NOTE Horizontal distances add to 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers on opposite sides of a tower each measure an angle of elevation to its top, and the distance between them is known. Explain why the tower's height cannot be found from one of the two right triangles alone.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

1913

QUESTION ID

S52-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Each triangle alone has two unknowns (h and its distance); the shared height and total distance give the needed second equation.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 40°, B = 65°, and a = 12. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

1915

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16.9

GRADING NOTE Rounding to 0.1 required. b = a sin B ÷ sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 52°, C = 70°, and c = 20. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

1916

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.8

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

True or false?

SOH-CAH-TOA applies directly to a triangle with no right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

1917

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Those ratios are defined in right triangles; use the Law of Sines or Cosines, or draw an altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

Which data allow the Law of Sines directly?

A. Two sides and the included angle

B. Two angles and any side

C. Three sides

D. Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

1918

QUESTION ID

S52-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Two angles and any side

GRADING NOTE AAS or ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

1919

QUESTION ID

S52-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The altitude creates two right triangles that share it, so h = b sin A = a sin B.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSG-C.A.1 • CIRCLES

Prove that all circles are similar

MAIN	Geometry main question	**HSG-C.A.1**
BACKWARD 1	A dilation multiplies distances by the scale factor	**HSG-SRT.A.1.b**
BACKWARD 2	Translate a point to another point	**8.G.A.3**
BACKWARD 3	A circle is all points at one distance from its center	**HSG-CO.A.1**
FORWARD 1	Find a single dilation that maps one circle onto another	**HSG-C.A.1**
FORWARD 2	Arcs of similar circles scale with the radius	**HSG-C.B.5**

SET 53 • HSG-C.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

1922

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation followed by a dilation maps C onto D, so they are similar.

GRADING NOTE REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r. Show the image of C is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: (1) the translation maps C onto the circle with center B and radius r (distances kept); (2) the dilation about B with factor s/r maps each point at distance r from B to a point at distance s on the same ray, so onto D; (3) every point of D is reached (the inverse dilation); a similarity transformation exists. Invalid: "circles all look alike"; only checking two specific circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center (1, 2) and radius 3. Circle C_2 has center (−4, 0) and radius 6. Describe a similarity transformation that maps C_1 onto C_2 and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

1923

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Translate 5 units left and 2 units down, then dilate about (−4, 0) with scale factor 2.

GRADING NOTE

REQUIRED WORK: Give the translation that moves the centers together and the dilation (center and factor).

Any valid proof (paragraph, two-column, or flow). Essential: the translation (x, y) → (x − 5, y − 2) moves the center to (−4, 0) keeping the radius 3; the dilation about (−4, 0) by 2 doubles every distance from the center, so radius 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

In the proof that all circles are similar, why must the scale factor be the ratio of the radii (s/r)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

1924

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

A dilation about the center multiplies every distance from the center by the factor, so the radius r must become s: k · r = s.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

1925

QUESTION ID

S53-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The dilation about the common center with factor s/r maps the first circle onto the second.

GRADING NOTE REQUIRED WORK: Name the transformation and its center and factor.

Any valid proof (paragraph, two-column, or flow). Essential: dilation about the shared center; factor s/r; each point at distance r goes to distance s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

True or false?

A circle can be similar to an ellipse that is not a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

1926

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A similarity keeps the ratio of all distances from the center, so the image of a circle is a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

1928

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

True or false?

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

1929

QUESTION ID

S53-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which factor maps a circle of radius 4 about its center to one of radius 10?

A. 2.5

B. 0.4

C. 6

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

1930

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

1931

QUESTION ID

S53-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A dilation can turn a circle into a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

1932

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

1934

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 units left and 2 units down

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

True or false?

A translation keeps the radius of a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

1935

QUESTION ID

S53-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

A. (6, 2)

B. (0, 4)

C. (0, 3)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

1936

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 4)

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

1937

GRADING NOTE Any equivalent form.

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 units right and 5 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A translation changes the size of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

1938

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Every point of a circle is the same distance from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

1940

QUESTION ID

S53-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

1941

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

A circle is completely determined by

A. its color

B. one point on it

C. its center only

D. its center and radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

1942

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. its center and radius

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

1943

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are the same circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

True or false?

A circle's radius can be different in different directions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

1944

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The definition requires one distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

1946

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−6, 0)

GRADING NOTE Form required: an ordered pair. The center (c, 0) satisfies 6 = c + 2(0 − c), so c = −6; check: (0, 0) → −6 + 2(0 + 6) = 6, and the radius 2 → 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

Circle P has center (0, 0) and radius 2. Circle Q has center (6, 0) and radius 4. What is the scale factor of a dilation that maps P onto Q?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

1947

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

True or false?

Two circles with different radii can always be mapped onto each other by one dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

1948

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the center on the line of centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

Circles with equal radii and different centers are related by

A. nothing

B. a translation (no dilation needed)

C. a dilation with factor 2

D. a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

1949

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a translation (no dilation needed)

GRADING NOTE Same size: congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

1950

QUESTION ID

S53-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Center (0, 0), factor 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

1950

SET 53 • HSG-C.A.1

FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

1952

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 : 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

1953

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

True or false?

For a fixed central angle, the arc length is the same in every circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

1954

QUESTION ID

S53-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Arc length is proportional to the radius; s/r is what stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

A. 2.4

B. 15

C. 30

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

1955

QUESTION ID

S53-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15

GRADING NOTE Multiply by 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

1956

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Since arc length is proportional to radius for a fixed angle, arc/radius depends only on the angle; that number is the angle in radians.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

Relate inscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Base angles of an isosceles triangle are congruent	**HSG-CO.C.10**
BACKWARD 2	The exterior angle theorem	**HSG-CO.C.10**
BACKWARD 3	A central angle and its arc have the same measure	**4.MD.C.5.a**
FORWARD 1	Prove the inscribed angle theorem when the center is inside or out…	**HSG-C.A.2**
FORWARD 2	Inscribed angles on the same arc, and arc applications	**HSG-C.A.2**

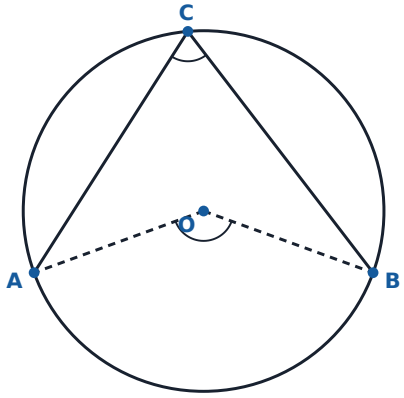
SET 54 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.



m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

1959

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

55°

GRADING NOTE Any equivalent form. An inscribed angle is half the central angle on the same arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

1960

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

Arc PR = 76°;
∠POR = 76°

GRADING NOTE Form required: both. The central angle equals its arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

1961

QUESTION ID

S54-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠BAC = ½ m∠BOC

GRADING NOTE REQUIRED WORK: Use isosceles triangle AOB (OA = OB) and the exterior angle theorem at O.

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii), so ∠OAB ≅ ∠OBA; ∠BOC is an exterior angle of △AOB, so m∠BOC = m∠OAB + m∠OBA = 2m∠BAC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

A. 23.5°

B. 94°

C. 47°

D. 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

1962

QUESTION ID

S54-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 94°

GRADING NOTE Double the inscribed angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

True or false?

An inscribed angle and a central angle that intercept the same arc have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

1963

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The inscribed angle is half the central angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In △AOB, OA = OB and m∠OAB = 35°. Find m∠OBA and m∠AOB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

1965

QUESTION ID
S54-B1-Q1

TYPE
Short answer

CORRECT ANSWER
35° and 110°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Two radii of a circle and the chord joining their ends form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

1966

QUESTION ID

S54-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The radii are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

Isosceles triangle with vertex angle 80°. Each base angle?

A. 100°

B. 50°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

1967

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50°

GRADING NOTE (180 − 80)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △OPQ with OP = OQ, m∠P = 2x and m∠Q = 50°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

1968

QUESTION ID

S54-B1-Q4

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

True or false?

In an isosceles triangle, the vertex angle always equals a base angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

1969

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

1971

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

True or false?

An exterior angle equals the sum of the two remote interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

1972

QUESTION ID

S54-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Exterior angle 100°, one remote interior angle 45°. The other?

A. 45°

B. 80°

C. 55°

D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

1973

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 55°

GRADING NOTE 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

1974

QUESTION ID

S54-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

True or false?

An exterior angle equals the adjacent interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

1975

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

ANSWER KEY Reduced copy of the question page, with its answer

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

1977

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

True or false?

The measure of a minor arc equals its central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

1978

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of arc measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc measures 70°. Its central angle measures

A. 70°

B. 140°

C. 290°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

1979

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 70°

GRADING NOTE Equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

1980

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

All arcs of 60° have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

1981

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length depends on the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC has the center O inside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

1983

QUESTION ID

S54-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m∠AOC

GRADING NOTE REQUIRED WORK: Split ∠ABC and ∠AOC along BD and apply the diameter case twice, then add.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ABD = ½m∠AOD and m∠DBC = ½m∠DOC (diameter case); add: m∠ABC = ½(m∠AOD + m∠DOC) = ½m∠AOC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC has the center O outside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

1984

QUESTION ID

S54-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m∠AOC

GRADING NOTE

REQUIRED WORK: Use the diameter case twice and SUBTRACT.

Any valid proof (paragraph, two-column, or flow). Essential: with D such that A and C are on the same side of BD: m∠DBC = ½m∠DOC and m∠DBA = ½m∠DOA; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

True or false?

The inscribed angle theorem needs three cases for a complete proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

1985

QUESTION ID

S54-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Center on a side, inside, or outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

In the proof of the inscribed angle theorem for the case where the center lies inside the inscribed angle, a diameter splits the angle in two, and the two diameter-case results are

A. averaged

B. multiplied

C. subtracted

D. added

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

1986

QUESTION ID

S54-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. added

GRADING NOTE The angle is the sum of two pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can't the proof of the inscribed angle theorem stop after the case in which one side of the angle is a diameter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

1987

QUESTION ID
S54-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Most inscribed angles do not have a diameter as a side; the other positions must be covered too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

1989

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

41°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two inscribed angles that intercept the same arc are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

1990

QUESTION ID

S54-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

They are congruent.

GRADING NOTE REQUIRED WORK: Each is half the same central angle.

Any valid proof (paragraph, two-column, or flow). Essential: each equals ½ of the central angle on that arc (inscribed angle theorem).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

True or false?

An inscribed angle that intercepts a 200° arc measures 200°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

1991

QUESTION ID

S54-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the arc: 100°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

A. 30°

B. 65°

C. 15°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

1992

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 65°

GRADING NOTE Angle formed by two chords = half the sum of the intercepted arcs: (80 + 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

1993

QUESTION ID

S54-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 3x + 10 = 55.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

SET 55 • HSG-C.A.2 • CIRCLES

Inscribed angles on a diameter are right angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Isosceles triangles from radii	**HSG-CO.C.10**
BACKWARD 3	Triangle angle sum	**8.G.A.5**
FORWARD 1	Use the converse to find a circle through a right triangle	**HSG-C.A.2**
FORWARD 2	Use the diameter property to construct tangents from an external…	**HSG-C.A.4**

SET 55 • HSG-C.A.2

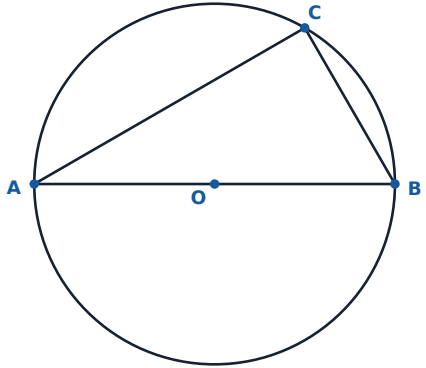
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

1996

QUESTION ID

S55-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACB = 90°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem (the intercepted arc is 180°), or the two isosceles triangles AOC and BOC.

Any valid proof (paragraph, two-column, or flow). Essential (theorem): ∠ACB intercepts a semicircle, 180°, so it measures 90°. Alternative: OA = OC = OB, so △AOC and △BOC are isosceles; with base angles x and y, the angle sum of △ABC gives 2x + 2y = 180°, so x + y = 90° = m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

1997

QUESTION ID

S55-M-Q2

TYPE

Short answer

CORRECT ANSWER

AB = 10; radius 5

GRADING NOTE Form required: both. ∠C = 90°, so AB = √(36 + 64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is on the circle with m∠CAB = 34°. Find m∠CBA.

m∠CBA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

1998

QUESTION ID
S55-M-Q3

TYPE
Short answer

CORRECT ANSWER
56°

GRADING NOTE Any equivalent form. ∠C = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

Triangle PQR is inscribed in a circle with PR a diameter. Which statement must be true?

A. PQ = QR

B. PR = PQ

C. ∠P = 90°

D. ∠Q = 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

1999

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠Q = 90°

GRADING NOTE The angle opposite the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every triangle inscribed in a circle is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

2000

QUESTION ID

S55-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when one side is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

2002

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle is half its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

2003

QUESTION ID

S55-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A semicircle measures

A. 360°

B. 180°

C. 90°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

2004

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180°

GRADING NOTE Half the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

2005

QUESTION ID

S55-B1-Q4

TYPE

Short answer

CORRECT ANSWER

50°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

2006

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

2008

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any two radii and the chord between them form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

2009

QUESTION ID

S55-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

2009

ANSWER KEY Reduced copy of the question page, with its answer

In △OBC with OB = OC, m∠BOC = 100°. m∠OBC =

A. 80°

B. 40°

C. 100°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2010

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40°

GRADING NOTE (180 − 100)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2010

ANSWER KEY Reduced copy of the question page, with its answer

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

2011

QUESTION ID

S55-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Base angles of an isosceles triangle are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A radius is always longer than a chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

2012

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Chords can be longer (up to the diameter).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

2014

QUESTION ID

S55-B3-Q1

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The acute angles of a right triangle add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

2015

QUESTION ID

S55-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 180 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2x + 2y = 180$. Then $x + y =$

A. 45

B. 360

C. 90

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2016

QUESTION ID

S55-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 90

GRADING NOTE Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2016

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

2017

QUESTION ID

S55-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A triangle can have angles 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

2018

QUESTION ID

S55-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

2018

SET 55 • HSG-C.A.2

FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2020

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Center (4, 3), the midpoint of AB; radius 5.

GRADING NOTE Form required: both. The hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2020

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, m∠C = 90°. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

2021

QUESTION ID

S55-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

MC = MA = MB, so C is on the circle with center M and diameter AB.

GRADING NOTE REQUIRED WORK: Let M be the midpoint of AB and show MC = MA (for example, by completing a rectangle).

Any valid proof (paragraph, two-column, or flow). Essential: reflect C through M to D; ACBD is a parallelogram with a right angle, so a rectangle; its diagonals are congruent and bisect each other, so MC = MA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2022

QUESTION ID

S55-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2022

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

A. 6.5

B. 8.5

C. 6

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

2023

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6.5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

2023

ANSWER KEY Reduced copy of the question page, with its answer

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

2024

QUESTION ID

S55-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A diameter of the board.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

2026

QUESTION ID

S55-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the midpoint M of OP; draw the circle with center M through O and P; it meets circle O at T_1 and T_2; lines PT_1 and PT_2 are tangents.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular bisector of OP, the circle on diameter OP, and both tangent lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct tangents from P to circle O, a circle with diameter OP is drawn; it meets circle O at T_1 and T_2. Why are lines PT_1 and PT_2 tangent to circle O?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

2027

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

T_1 is on the circle with diameter OP, so $\angle OT_1P = 90°$; a line perpendicular to a radius at its endpoint is tangent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

2027

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From a point outside a circle there is exactly one tangent line to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

2028

QUESTION ID
S55-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE There are two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, a circle with diameter OP is drawn. That circle is used because

A. it passes through the center only

B. every angle inscribed in it on OP is a right angle

C. it has the same radius as circle O

D. it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

2029

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. every angle inscribed in it on OP is a right angle

GRADING NOTE Right angle at the tangent point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

2030

QUESTION ID

S55-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

A radius is perpendicular to the tangent at the point of tangency

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the Pythagorean Theorem and its converse	**8.G.B.7**
BACKWARD 2	The perpendicular is the shortest segment from a point to a line	**7.G.A.2**
BACKWARD 3	A tangent touches a circle at exactly one point	**HSG-CO.A.1**
FORWARD 1	Tangent segments from an external point are congruent	**HSG-C.A.2**
FORWARD 2	Write the equation of a tangent line in the coordinate plane	**HSG-GPE.B.5**

SET 56 • HSG-C.A.2

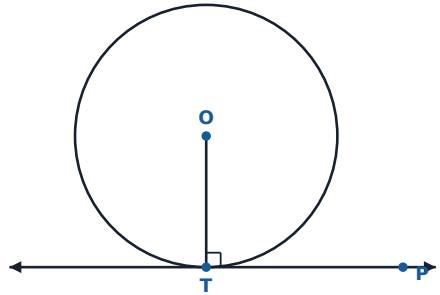
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

2033

QUESTION ID

S56-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

OT $\perp \ell$

GRADING NOTE REQUIRED WORK: Suppose OT is not perpendicular; drop the perpendicular from O to ℓ at Q ≠ T; show a second point of ℓ is on the circle or inside it.

Any valid proof (paragraph, two-column, or flow). Essential: if OQ $\perp \ell$ with Q ≠ T, then OQ < OT (the perpendicular is the shortest segment, as the hypotenuse OT is longest in right △OQT), so Q is inside the circle, and then ℓ would cross the circle twice, contradicting tangency. Invalid: "it looks perpendicular".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

2034

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
13

GRADING NOTE Any equivalent form. ∠OTP = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

2035

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Segment PT touches circle O at T. OT = 7, PT = 24, OP = 26. Is PT tangent to the circle?

A. Yes, because 26 > 24

B. Yes, because it touches at T

C. No, because 7 < 24

D. No, because $7^2 + 24^2 \neq 26^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

2036

QUESTION ID
S56-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. No, because $7^2 + 24^2 \neq 26^2$

GRADING NOTE Tangency requires ∠OTP = 90°, but 49 + 576 = 625 ≠ 676.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

2037

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

2039

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

2040

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

True or false?

A triangle with sides 7, 24, 25 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

2041

QUESTION ID

S56-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 49 + 576 = 625.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

A. Yes

B. No

C. Only if 11 is a leg

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

2042

QUESTION ID

S56-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. No

GRADING NOTE 36 + 64 ≠ 121.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The Pythagorean Theorem applies to any triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

2043

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The shortest segment from a point to a line is perpendicular to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2045

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any other segment is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2046

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

O is 4 cm from line ℓ (perpendicular distance). Another segment from O to ℓ could be

A. 4 cm only

B. 5 cm

C. 2 cm

D. 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2047

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 cm

GRADING NOTE At least 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2048

QUESTION ID

S56-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Each slanted segment is the hypotenuse of a right triangle with the perpendicular as a leg.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The distance from a point to a line is measured along any segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2049

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Along the perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

A tangent line meets the circle at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

2051

QUESTION ID
S56-B3-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line meets a circle at two points. It is a

A. secant

B. radius

C. diameter

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

2052

QUESTION ID

S56-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. secant

GRADING NOTE Two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

2053

QUESTION ID

S56-B3-Q3

TYPE

Short answer

CORRECT ANSWER

No; it would cross the circle twice.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

2054

QUESTION ID

S56-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The point of tangency

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

True or false?

A tangent can pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

2055

QUESTION ID

S56-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Then it would meet the circle twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSG-C.A.2

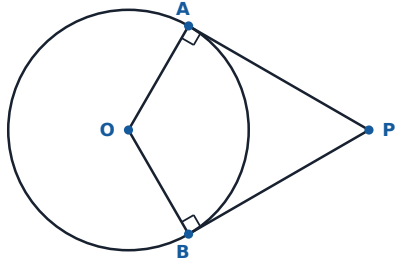
FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

2057

QUESTION ID

S56-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PA = PB

GRADING NOTE REQUIRED WORK: Use the right angles at A and B, OA = OB, and the shared hypotenuse OP (HL, or the Pythagorean Theorem).

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90°; OA = OB; OP shared; HL (or PA = $\sqrt{(OP^2 - r^2)}$ = PB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

2058

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

True or false?

The segment from P to the center bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

2059

QUESTION ID
S56-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

A. equal

B. parallel

C. unequal

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

2060

QUESTION ID

S56-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal

GRADING NOTE Same external point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

2060

ANSWER KEY Reduced copy of the question page, with its answer

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

2061

QUESTION ID

S56-F1-Q5

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form. AB + CD = BC + DA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find an equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

2063

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Any equivalent form, such as 3x + 4y = 25. The radius has slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

2064

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2x − 3y = 13

GRADING NOTE Any equivalent form. Radius slope −3/2; tangent slope 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

True or false?

A tangent line has the same slope as the radius to its point of tangency.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

2065

QUESTION ID
S56-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its slope is the negative reciprocal (when neither line is vertical or horizontal).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The tangent to $x^2 + y^2 = 16$ at (0, 4) is

A. $x = 0$

B. $y = x + 4$

C. $x = 4$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

2066

QUESTION ID

S56-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 4

GRADING NOTE The radius is vertical, so the tangent is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

2067

QUESTION ID

S56-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: the distance from (2, 3) to the line y = 7 is 4, the radius.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

Relate circumscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	A tangent is perpendicular to the radius at the point of tangency	**HSG-C.A.2**
BACKWARD 2	The angles of a quadrilateral add to 360°	**8.G.A.5**
BACKWARD 3	Supplementary angles add to 180°	**7.G.B.5**
FORWARD 1	Use the circumscribed angle with arcs, including major arcs	**HSG-C.A.2**
FORWARD 2	Angles formed by secants and tangents outside a circle	**HSG-C.A.2**

SET 57 • HSG-C.A.2

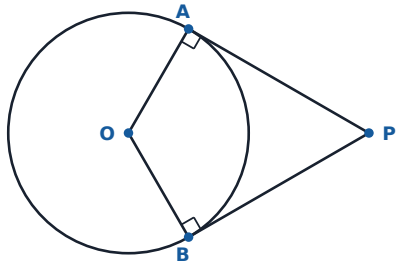
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Prove that m∠APB + m∠AOB = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

2070

QUESTION ID

S57-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠APB + m∠AOB = 180°

GRADING NOTE REQUIRED WORK: Use the right angles at A and B and the angle sum of quadrilateral OAPB.

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90° (radius ⊥ tangent); the angles of quadrilateral OAPB add to 360°; so m∠APB + m∠AOB = 360° − 180° = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

2071

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

∠AOB = 128°; arc AB = 128°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Minor arc AB measures 100°. Find the circumscribed angle formed by the tangents at A and B.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

2072

QUESTION ID
S57-M-Q3

TYPE
Short answer

CORRECT ANSWER
80°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

A circumscribed angle measures 90°. The quadrilateral formed by the center, the two tangent points, and the vertex is a

A. kite that is not a square

B. trapezoid

C. rhombus that is not a square

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2073

QUESTION ID
S57-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. square

GRADING NOTE All four angles are 90°, so OAPB is a rectangle; its opposite sides give OA = PB, and with PA = PB and OA = OB all four sides are equal, so it is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2073

True or false?

A circumscribed angle and its central angle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

2074

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

The radius to the point of tangency is perpendicular to the tangent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

2076

QUESTION ID

S57-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

2077

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

A. 4

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

2078

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE At A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

2079

QUESTION ID

S57-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angle between a tangent and a radius can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

2080

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90° at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

2082

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

True or false?

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

2083

QUESTION ID

S57-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 × 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a quadrilateral are right angles. The other two must

A. each be 90°

B. add to 360°

C. add to 180°

D. add to 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

2084

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. add to 180°

GRADING NOTE 360 − 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

2085

QUESTION ID

S57-B2-Q4

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angles of a quadrilateral add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

2086

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

2086

SET 57 • HSG-C.A.2

BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

2088

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

128°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

True or false?

Supplementary angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

2089

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 3x are supplementary. x =

A. 60°

B. 90°

C. 45°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2090

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2090

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

2091

QUESTION ID

S57-B3-Q4

TYPE

Short answer

CORRECT ANSWER

acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

2092

QUESTION ID

S57-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

The circumscribed angle at P is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

2094

QUESTION ID
S57-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Minor arc 140°; major arc 220°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

Show that a circumscribed angle equals ½(major arc − minor arc).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

2095

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

If the minor arc is m, the major arc is 360 − m and the angle is 180 − m = ½((360 − m) − m).

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A circumscribed angle can measure more than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

2096

QUESTION ID

S57-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 180° minus the minor arc, so it is less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P make a 70° angle. An inscribed angle intercepting the minor arc AB measures

A. 110°

B. 70°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2097

QUESTION ID

S57-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 55°

GRADING NOTE Minor arc = 110°, inscribed angle = 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The minor arc between two tangent points is x°, and the circumscribed angle is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

2098

QUESTION ID
S57-F1-Q5

TYPE
Short answer

CORRECT ANSWER
100

GRADING NOTE Any equivalent form. x + x − 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

2100

QUESTION ID

S57-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠P = ½(arc BD − arc AC)

GRADING NOTE REQUIRED WORK: Draw chord BC; use the exterior angle theorem in triangle PBC and the inscribed angle theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BCD is exterior to △PBC, so m∠BCD = m∠P + m∠PBC; m∠BCD = ½ arc BD and m∠PBC = ½ arc AC; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

2101

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

True or false?

The angle outside a circle formed by a tangent and a secant is half the sum of the intercepted arcs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

2102

QUESTION ID

S57-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the difference; half the sum is for two chords meeting inside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

A. 110°

B. 55°

C. 210°

D. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

2103

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 55°

GRADING NOTE (160 − 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

2104

QUESTION ID
S57-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The tangent case is the limit where each secant's two intersection points merge; the arcs are the major and minor arcs.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

Relationships among radii and chords

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Right triangles: Hypotenuse-Leg and the Pythagorean Theorem	**HSG-SRT.B.5**
BACKWARD 2	The perpendicular bisector theorem	**HSG-CO.C.9**
BACKWARD 3	Radii of a circle are equal	**HSG-CO.A.1**
FORWARD 1	Locate the center of a circle from chords	**HSG-C.A.2**
FORWARD 2	Find chord lengths and central angles with trigonometry	**HSG-SRT.C.8**

SET 58 • HSG-C.A.2

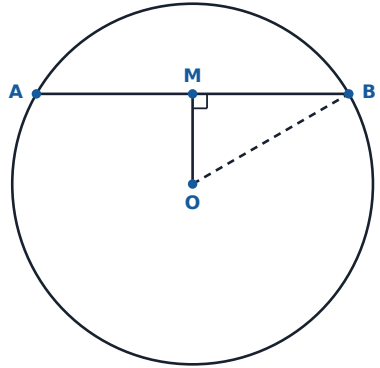
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

2107

QUESTION ID

S58-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AM = MB

GRADING NOTE REQUIRED WORK: Prove △OMA ≅ △OMB (HL, using OA = OB).

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii); OM shared; right angles at M; HL (or the Pythagorean Theorem: AM = √(r² − OM²) = MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

2108

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
24

GRADING NOTE Any equivalent form. Half-chord $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

2109

QUESTION ID
S58-M-Q3

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

Two chords of a circle are each 7 units from the center. Which is true?

A. The chords are perpendicular

B. The chords are congruent

C. The chords are parallel

D. The chords meet at the center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2110

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The chords are congruent

GRADING NOTE Equidistant chords are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2110

True or false?

The perpendicular bisector of any chord passes through the center of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

2111

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The center is equidistant from the chord's endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

2113

QUESTION ID
S58-B1-Q1

TYPE
Short answer

CORRECT ANSWER
HL

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

2114

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

True or false?

In circle O, OM is perpendicular to chord AB at M. The radii OA and OB are congruent hypotenuses of right triangles OMA and OMB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

2115

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE OA = OB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OM ⊥ AB in circle O. △OMA and △OMB share

A. nothing

B. OM

C. AB

D. OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

2116

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. OM

GRADING NOTE Common leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

True or false?

HL works for any triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

2117

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

2119

QUESTION ID

S58-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

2120

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

OA and OB are radii.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

The center of a circle is equidistant from the endpoints of each of its chords. So the center lies on

A. the perpendicular bisector of every chord

B. a tangent

C. the circle

D. every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

2121

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the perpendicular bisector of every chord

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two chords' perpendicular bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

2122

QUESTION ID

S58-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The center of the circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of a chord can miss the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

2123

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It always passes through it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

2125

QUESTION ID

S58-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

True or false?

All radii of one circle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

2126

QUESTION ID

S58-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

AB is a chord of circle O, and OA = 5. OB =

A. unknown

B. 10

C. 2.5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

2127

QUESTION ID

S58-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE Radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

2128

QUESTION ID

S58-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A diameter, 12

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A chord always passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

2129

QUESTION ID

S58-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

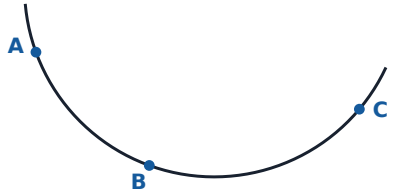
SET 58 • HSG-C.A.2

FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

2131

QUESTION ID

S58-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw chords AB and BC; construct their perpendicular bisectors; they meet at the center.

GRADING NOTE Grade the construction marks, not only the final figure: two chords and both perpendicular bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

2132

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Form required: an ordered pair. Perpendicular bisectors x = 4 and y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

True or false?

Three points not on one line determine exactly one circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

2133

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The perpendicular bisectors meet in one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

Why can't three points on one line lie on a circle?

A. Three points on a line always lie on a circle

B. Their perpendicular bisectors are parallel, so no center exists

C. Only two points can lie on a circle

D. They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

2134

QUESTION ID

S58-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Their perpendicular bisectors are parallel, so no center exists

GRADING NOTE Parallel bisectors never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

2135

QUESTION ID
S58-F1-Q5

TYPE
Short answer

CORRECT ANSWER

Mark three points on the edge, construct the perpendicular bisectors of two chords to find the center, then measure to the edge.

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

2137

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12.9

GRADING NOTE Rounding to 0.1 required. Half the chord = 10 sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

2138

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

73.7°

GRADING NOTE Rounding to 0.1° required. Half-angle $\sin^{-1}(6/10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A chord's length is 2r sin θ for central angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

2139

QUESTION ID

S58-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2r sin(θ/2), from the right triangle with half the chord.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Central angle 60°, radius 9. Chord length?

A. 4.5

B. 18

C. 9

D. 9√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

2140

QUESTION ID

S58-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

GRADING NOTE $2 \cdot 9 \cdot \sin 30° = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

2141

QUESTION ID

S58-F2-Q5

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. Equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

Construct the inscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct an angle bisector	**HSG-CO.D.12**
BACKWARD 2	Construct a perpendicular from a point to a line	**HSG-CO.D.12**
BACKWARD 3	Points on an angle bisector are equidistant from the sides	**HSG-SRT.B.5**
FORWARD 1	Prove that the incenter is equidistant from the sides	**HSG-C.A.3**
FORWARD 2	Locate a point equidistant from three lines in a design	**HSG-MG.A.3**

SET 59 • HSG-C.A.3

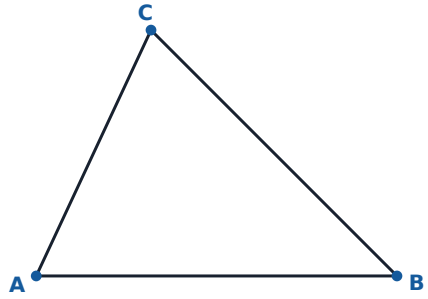
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

2144

QUESTION ID

S59-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles (for example ∠A and ∠B); they meet at the incenter I; construct the perpendicular from I to one side, meeting it at D; draw the circle with center I and radius ID.

GRADING NOTE Grade the construction marks, not only the final figure: two angle bisector constructions, the perpendicular from I to a side, and the circle touching all three sides.

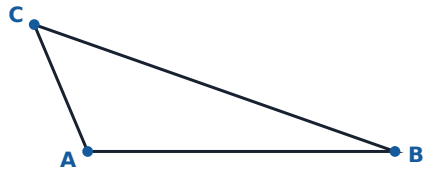
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

2145

QUESTION ID

S59-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps: two angle bisectors meet at I (inside the triangle), perpendicular from I to a side gives the radius, then the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

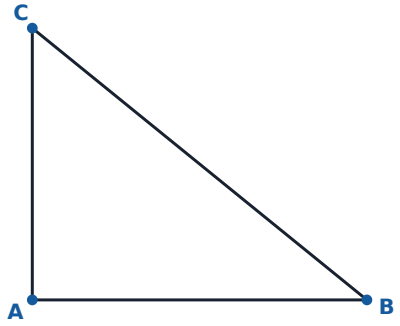
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

2146

QUESTION ID

S59-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Two angle bisectors, the perpendicular from their intersection to a side, and the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

Write the steps for constructing the inscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

2147

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

1. Construct the bisectors of two angles; label their intersection I. 2. Construct the perpendicular from I to one side; label the foot D. 3. Draw the circle with center I through D.

GRADING NOTE Any equivalent steps. The radius must come from a PERPENDICULAR to a side, not from I to a vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

Why is the radius of the inscribed circle measured along a perpendicular from the incenter to a side?

A. The radius goes to a vertex

B. Any segment from I works

C. The circle must be tangent to the side, and a tangent is perpendicular to the radius

D. The perpendicular is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

2148

QUESTION ID

S59-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The circle must be tangent to the side, and a tangent is perpendicular to the radius

GRADING NOTE Tangent ⊥ radius at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

2150

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Three (one at the vertex, then two equal arcs from the crossing points)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

True or false?

An angle bisector construction ends with a ray from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

2151

QUESTION ID

S59-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Through the crossing of the equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

The incenter is where the

A. medians meet

B. perpendicular bisectors meet

C. angle bisectors meet

D. altitudes meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

2152

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. angle bisectors meet

GRADING NOTE Equidistant from sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

2153

QUESTION ID

S59-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The third bisector passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

True or false?

Bisecting the sides locates the incenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

2154

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That locates the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

2156

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular from the incenter to the side

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

True or false?

The perpendicular from a point to a line is the shortest segment to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

2157

QUESTION ID

S59-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

A. Measure AB

B. Draw a circle at A

C. Bisect ∠I

D. Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

2158

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Arcs of equal radius from those two points, crossing on the other side

GRADING NOTE Then draw the line through I and that crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

2159

QUESTION ID

S59-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Vertices are farther than the sides; the circle must touch the sides, not pass through vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

True or false?

The incircle passes through the vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

2160

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

True or false?

A point on the bisector of ∠A is equidistant from the two sides of ∠A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

2162

QUESTION ID

S59-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

2163

QUESTION ID
S59-B3-Q2

TYPE
Short answer

CORRECT ANSWER
3 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

A. to the nearest vertex

B. along the perpendicular

C. along the bisector

D. along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

2164

QUESTION ID

S59-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. along the perpendicular

GRADING NOTE Shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

2165

QUESTION ID

S59-B3-Q4

TYPE

Short answer

CORRECT ANSWER

It is on all three angle bisectors, and each bisector point is equidistant from that angle's sides.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

True or false?

The incenter is equidistant from the three vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

2166

QUESTION ID
S59-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That is the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

2168

QUESTION ID

S59-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

I is equidistant from AB, BC, CA and lies on the bisector of ∠C.

GRADING NOTE REQUIRED WORK: Use the angle bisector distance property twice, then its converse at C.

Any valid proof (paragraph, two-column, or flow). Essential: on the bisector of ∠A: d(I, AB) = d(I, AC); on the bisector of ∠B: d(I, AB) = d(I, BC); so d(I, AC) = d(I, BC); the converse puts I on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

2169

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Area = r · (perimeter ÷ 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

True or false?

The incenter of an obtuse triangle lies outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

2170

QUESTION ID

S59-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The incenter is always inside; the circumcenter is the one that can be outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

2170

A 3-4-5 right triangle has inradius

A. 2.5

B. 1

C. 1.5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

2171

QUESTION ID

S59-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE Area 6 = r · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

2172

QUESTION ID

S59-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Joining I to the vertices splits the triangle into three triangles with heights r on bases a, b, c: area = ½r(a + b + c) = r · s.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

2174

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 m

GRADING NOTE Any equivalent form. s = 21, r = 84/21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

2175

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A larger circle would cross some side; the incircle touches all three sides.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

True or false?

A sprinkler equidistant from three straight fences forming a triangle should be at the circumcenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

2176

QUESTION ID

S59-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the incenter (equidistant from lines); the circumcenter is equidistant from the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

To be equidistant from three towns (points), use the

A. orthocenter

B. centroid

C. incenter

D. circumcenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

2177

QUESTION ID

S59-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. circumcenter

GRADING NOTE Points, not lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

2178

QUESTION ID
S59-F2-Q5

TYPE
Short answer

CORRECT ANSWER
√3

GRADING NOTE Form required: exact. Area 9√3, s = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

Construct the circumscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 2	Points on the perpendicular bisector are equidistant from the end…	**HSG-CO.C.9**
BACKWARD 3	Triangle types: acute, right, obtuse	**4.G.A.2**
FORWARD 1	Find the circumcenter and radius with coordinates	**HSG-GPE.B.5**
FORWARD 2	Find the circumradius with trigonometry (extended Law of Sines)	**HSG-SRT.D.10**

SET 60 • HSG-C.A.3

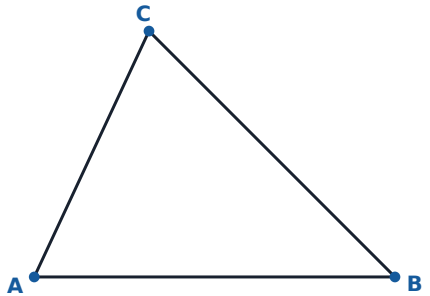
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

2181

QUESTION ID

S60-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisectors of two sides; they meet at O; draw the circle with center O through A (it also passes through B and C).

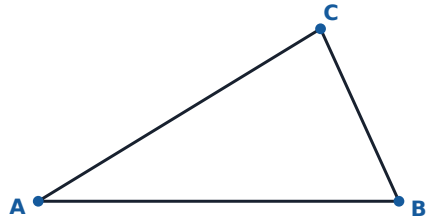
GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions and the circle through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

2182

QUESTION ID

S60-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps; the circumcenter is outside the triangle.

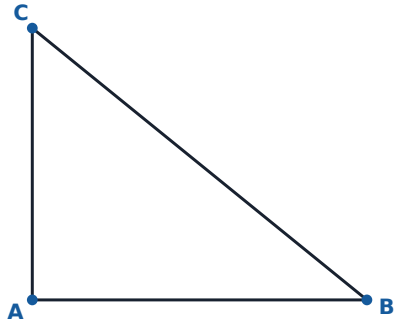
GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

2183

QUESTION ID

S60-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisectors meet at the midpoint of hypotenuse BC; circle centered there through A.

GRADING NOTE Grade the construction marks, not only the final figure: constructions and the circle; the center is on BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

Write the steps for constructing the circumscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

2184

QUESTION ID

S60-M-Q4

TYPE

Short answer

CORRECT ANSWER

1. Construct the perpendicular bisectors of two sides. 2. Label their intersection O. 3. Draw the circle with center O through one vertex.

GRADING NOTE Any equivalent steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

True or false?

The circumcenter of an obtuse triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

2185

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It lies outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

2187

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector construction uses equal arcs from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

2188

QUESTION ID

S60-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcenter is where the

A. angle bisectors meet

B. sides meet

C. perpendicular bisectors of the sides meet

D. medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

2189

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. perpendicular bisectors of the sides meet

GRADING NOTE Equidistant from vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

2190

QUESTION ID

S60-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The arcs in a perpendicular bisector construction can have any two different radii.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

2191

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

2193

QUESTION ID

S60-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

O is on the perpendicular bisectors of AB and BC. Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

2194

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

OA = OB = OC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcircle's radius is

A. the distance from O to any vertex

B. the longest side

C. the distance from O to a side

D. half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

2195

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the distance from O to any vertex

GRADING NOTE Through the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

2196

QUESTION ID

S60-B2-Q4

TYPE

Short answer

CORRECT ANSWER

On the perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The circumcenter is equidistant from the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

2197

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE From the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

2199

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Obtuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A right triangle has one 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

2200

QUESTION ID

S60-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

2200

Where is the circumcenter of a right triangle?

A. Inside

B. Outside

C. On the hypotenuse (its midpoint)

D. At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

2201

QUESTION ID

S60-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. On the hypotenuse (its midpoint)

GRADING NOTE Hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

2202

QUESTION ID

S60-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An acute triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

2203

QUESTION ID

S60-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All angles are acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

2205

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 1); radius √10

GRADING NOTE Form required: both. Bisector of AB: x = 3; of AC: y − 2 = −½(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

2205

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

2206

QUESTION ID
S60-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Form required: an ordered pair. All three are 2 from the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

True or false?

The circumcenter is the intersection of two perpendicular bisector equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

2207

QUESTION ID
S60-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Solve the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

A. 5

B. 10

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

2208

QUESTION ID

S60-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

2209

QUESTION ID

S60-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Verify it is the same distance from all three vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In a triangle, a = 10 and A = 30°. Find the circumradius R using a/sin A = 2R.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

2211

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. 10 / 0.5 = 20 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

2212

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$2\sqrt{3}$

GRADING NOTE Form required: exact. $6/\sin 60° = 4\sqrt{3} = 2R$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

True or false?

In any triangle, a/sin A equals the radius of the circumcircle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

2213

QUESTION ID
S60-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It equals the diameter, 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

A triangle with a 90° angle opposite side 14 has circumradius

A. 3.5

B. 14

C. 28

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

2214

QUESTION ID

S60-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE 14/sin 90° = 14 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

Explain why a/sin A = 2R using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

2215

QUESTION ID

S60-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Draw the diameter from B; the inscribed angle at its far end equals A (same arc) and is in a right triangle with hypotenuse 2R, so sin A = a/(2R).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSG-C.A.3 • CIRCLES

Prove that opposite angles of an inscribed quadrilateral are supplementary

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	The inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Supplementary angles	**7.G.B.5**
BACKWARD 3	Arcs of a circle add to 360°	**4.MD.C.7**
FORWARD 1	Use the converse: decide whether a quadrilateral can be inscribed	**HSG-C.A.3**
FORWARD 2	Exterior angle of a cyclic quadrilateral	**HSG-C.A.3**

SET 61 • HSG-C.A.3

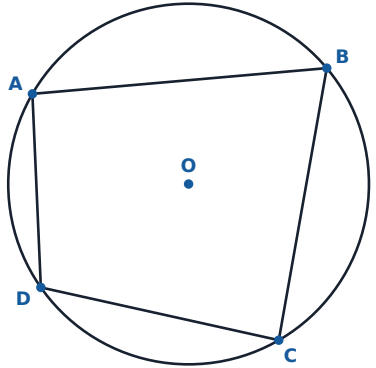
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in circle O. Prove that m∠A + m∠C = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

2218

QUESTION ID

S61-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠C = 180°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠A and ∠C intercept the two arcs BCD and BAD, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A = ½ arc BCD and m∠C = ½ arc BAD (inscribed angle theorem); the two arcs add to 360°; so the angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral PQRS is inscribed in a circle. Prove that ∠Q and ∠S are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

2219

QUESTION ID

S61-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠Q + m∠S = 180°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠Q and ∠S intercept arcs PSR and PQR, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: each angle is half its intercepted arc, and the two intercepted arcs make up the whole circle (360°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in a circle with m∠A = 72° and m∠B = 105°. Find m∠C and m∠D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

2220

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

m∠C = 108°, m∠D = 75°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrilateral can always be inscribed in a circle?

A. Any rhombus

B. Any trapezoid

C. Any parallelogram

D. Any rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

2221

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Any rectangle

GRADING NOTE Opposite angles 90° + 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

2221

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A parallelogram that is not a rectangle can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

2222

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its opposite angles are equal but not 90°, so they are not supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

2224

QUESTION ID
S61-B1-Q1

TYPE
Short answer

CORRECT ANSWER
110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle can intercept an arc greater than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

2225

QUESTION ID

S61-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then the angle is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle of 70° intercepts

A. 70°

B. 290°

C. 35°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

2226

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 140°

GRADING NOTE Double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

2227

QUESTION ID

S61-B1-Q4

TYPE

Short answer

CORRECT ANSWER

360°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals half the central angle only when it is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

2228

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

2230

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

True or false?

Two angles whose measures add to 180° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

2231

QUESTION ID

S61-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles 3x and 2x are supplementary. x =

A. 30°

B. 45°

C. 36°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

2232

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 36°

GRADING NOTE 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If ∠A = ∠C and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

2233

QUESTION ID

S61-B2-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

2234

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only if both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

2236

QUESTION ID

S61-B3-Q1

TYPE

Short answer

CORRECT ANSWER

180

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

True or false?

The arcs between four points on a circle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

2237

QUESTION ID

S61-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two arcs make a whole circle; one is 150°. The other is

A. 210°

B. 150°

C. 30°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

2238

QUESTION ID

S61-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 210°

GRADING NOTE 360 − 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

2238

ANSWER KEY Reduced copy of the question page, with its answer

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

2239

QUESTION ID

S61-B3-Q4

TYPE

Short answer

CORRECT ANSWER

180°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

True or false?

The arcs of a circle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

2240

QUESTION ID

S61-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

2242

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: A + C = 180° and B + D = 180°.

GRADING NOTE Form required: the conclusion with the check. (The converse is true: such a quadrilateral is cyclic.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

2242

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

2243

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: its other angles are 116°, and 64° + 116° = 180° for each pair of opposite angles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

True or false?

Every kite can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

2244

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only kites with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

A. 40

B. 50

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

2245

QUESTION ID

S61-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 40

GRADING NOTE 4x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed quadrilateral: ∠A = (5y − 10)°,
∠C = (3y + 30)°, A and C opposite. Find y and ∠A.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

2246

QUESTION ID
S61-F1-Q5

TYPE
Short answer

CORRECT ANSWER
y = 20; ∠A = 90°

GRADING NOTE Form required: both. 8y + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

2248

QUESTION ID

S61-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠BCE = m∠A

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the supplementary opposite angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠BCE = 180° − m∠BCD (linear pair); m∠A = 180° − m∠BCD (opposite angles); so equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In inscribed ABCD, ∠A = 77°. Side DC is extended past C to E. Find ∠BCE.

∠BCE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

2249

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

77°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

True or false?

An exterior angle of a cyclic quadrilateral equals the interior angle next to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

2250

QUESTION ID

S61-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is supplementary to the adjacent interior angle and equals the opposite one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed ABCD with ∠B = 100°. The exterior angle at D (extending CD past D) is

A. 50°

B. 180°

C. 100°

D. 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

2251

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 100°

GRADING NOTE Equals the opposite interior angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

ABCD is inscribed in a circle, and side DC is extended past C. Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

2252

QUESTION ID

S61-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It is supplementary to ∠C, and the angle supplementary to ∠C is its opposite angle, ∠A.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

Derive arc length from similarity and define radian measure

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Similar figures: lengths scale by the scale factor	**HSG-SRT.A.2**
BACKWARD 2	Circumference of a circle	**7.G.B.4**
BACKWARD 3	Find a fraction of a whole	**7.RP.A.3**
FORWARD 1	Convert between degrees and radians	**HSG-C.B.5**
FORWARD 2	Radian measure for angles of rotation and the unit circle	**HSF-TF.A.1**

SET 62 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle θ cuts off arc length s_1 on a circle of radius r_1 and arc length s_2 on a circle of radius r_2 (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

2255

QUESTION ID

S62-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$s_2 = (r_2/r_1)s_1$, so $s_1/r_1 = s_2/r_2$.

GRADING NOTE REQUIRED WORK: Use the dilation about the center with factor r_2/r_1: it maps the first circle to the second and the first arc to the second arc.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about the center with factor $k = r_2/r_1$ maps the first circle onto the second and the arc of angle θ onto the arc of angle θ; it multiplies all lengths, including arc length, by k; so $s_2 = k\, s_1$ and s/r is the same for both. This constant is θ in radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

2256

QUESTION ID
S62-M-Q2

TYPE
Short answer

CORRECT ANSWER
3 radians

GRADING NOTE Any equivalent form. θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

2257

QUESTION ID
S62-M-Q3

TYPE
Short answer

CORRECT ANSWER
15 m

GRADING NOTE Any equivalent form. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

2258

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

6π

GRADING NOTE Form required: exact value. 120° = 2π/3 radians; s = 9 · 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

Why is a full circle 2π radians?

A. There are 360 degrees

B. Every circle has radius 1

C. The circumference is 2πr, so s/r = 2π for the whole circle

D. π ≈ 3.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

2259

QUESTION ID

S62-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The circumference is 2πr, so s/r = 2π for the whole circle

GRADING NOTE Arc length ÷ radius for the whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

2259

SET 62 • HSG-C.B.5

BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

True or false?

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

2261

QUESTION ID
S62-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

2262

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

A. 20

B. 15

C. 8

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

2263

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

GRADING NOTE × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

2264

QUESTION ID

S62-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

True or false?

Arc lengths do not change under dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

2265

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

2267

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

2268

QUESTION ID

S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

A circle has diameter 8. Its circumference is

A. 64π

B. 4π

C. 16π

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

2269

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8π

GRADING NOTE πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

2270

QUESTION ID

S62-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$C = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

2271

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

2273

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

¼

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

2274

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

An arc of 45° is what fraction of the circumference?

A. 45/100

B. ⅛

C. 1/45

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

2275

QUESTION ID

S62-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅛

GRADING NOTE 45/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

True or false?

A 120° arc is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

2276

QUESTION ID

S62-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

True or false?

A 60° arc is 1/60 of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

2277

QUESTION ID

S62-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

2279

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5π/6

GRADING NOTE Form required: exact value. Multiply by π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

2280

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

135°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

True or false?

1 radian is about 3.14°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

2281

QUESTION ID

S62-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 radian ≈ 57.3°; π radians = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

A. 60°

B. 120°

C. 30°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

2282

QUESTION ID

S62-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 60°

GRADING NOTE 180/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

2283

QUESTION ID
S62-F1-Q5

TYPE
Short answer

CORRECT ANSWER
90°

GRADING NOTE Any equivalent form. θ = π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

2285

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. r = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel of radius 0.3 m turns through 10 radians. How far does a point on its rim travel?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

2286

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An angle of 6 radians is more than one full turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

2287

QUESTION ID
S62-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One full turn is $2\pi \approx 6.28$ radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

A point on the unit circle travels an arc of length π. The angle is

A. 1 radian

B. 2π radians

C. π°

D. π radians (180°)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

2288

QUESTION ID

S62-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π radians (180°)

GRADING NOTE θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

2289

QUESTION ID
S62-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Arc length is proportional to radius, so s/r is the same for every circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

Derive and use the formula for the area of a sector

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The radius is half the diameter	**7.G.B.4**
BACKWARD 3	Fractions of a circle in degrees and radians	**HSG-C.B.5**
FORWARD 1	Solve inverse sector problems	**HSG-C.B.5**
FORWARD 2	Areas of segments and composite circular regions	**HSG-MG.A.1**

SET 63 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2292

QUESTION ID

S63-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A = (θ/360)πr² for degrees; A = ½r²θ for radians.

GRADING NOTE REQUIRED WORK: Use the fraction of the whole circle that the sector is, times the circle's area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is θ/360 of the circle (central angles split the area in proportion); area = (θ/360)πr²; with θ radians, the fraction is θ/(2π), so A = (θ/(2π))πr² = ½r²θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a sector with a 60° central angle in a circle of radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2293

QUESTION ID
S63-M-Q2

TYPE
Short answer

CORRECT ANSWER
6π

GRADING NOTE Form required: exact value. (60/360)π(36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

2294

QUESTION ID
S63-M-Q3

TYPE
Short answer

CORRECT ANSWER
15

GRADING NOTE Any equivalent form. ½(25)(1.2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pizza with diameter 14 in. is cut into 8 equal slices. Find the area of one slice to the nearest tenth of a square inch.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

2295

QUESTION ID

S63-M-Q4

TYPE

Short answer

CORRECT ANSWER

19.2 in²

GRADING NOTE Rounding to 0.1 required. Radius 7 (not 14): (1/8)π(49).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sector has area 12π in a circle of radius 6. What is its central angle?

A. 240°

B. 12°

C. 120°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

2296

QUESTION ID

S63-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 120°

GRADING NOTE 12π/36π = ⅓ of 360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

2298

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

2299

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value. Use the radius 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$A = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

2300

QUESTION ID

S63-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

A circle has area 49π. Radius?

A. 14

B. 7

C. 24.5

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

2301

QUESTION ID

S63-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

GRADING NOTE $r^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Doubling the radius doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

2302

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the area by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

2304

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $A = \pi r^2$ gives an area four times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

2305

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2r)^2 = 4r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

A sector of a circle with diameter 20 uses r =

A. 10

B. 20

C. 5

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

2306

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2307

QUESTION ID

S63-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

True or false?

The radius and diameter can be used interchangeably in area formulas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

2308

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They differ by a factor of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

2310

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

2311

QUESTION ID

S63-B3-Q2

TYPE

Short answer

CORRECT ANSWER

¼

GRADING NOTE Any equivalent form. (π/2)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

A full circle measures

A. 360° or 2π radians

B. 180° or π radians

C. 360 radians

D. 2π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

2312

QUESTION ID

S63-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 360° or 2π radians

GRADING NOTE Both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

True or false?

A central angle of 1 radian is 1/(2π) of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

2313

QUESTION ID

S63-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fraction of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

True or false?

A 90° sector is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

2314

QUESTION ID

S63-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ¼.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2316

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $(40/360)\pi r^2 = 4\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

2317

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.6 radians

GRADING NOTE Any equivalent form. ½(100)θ = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

True or false?

If the radius is tripled and the angle stays the same, the sector area is tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

2318

QUESTION ID

S63-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area depends on r^2, so it is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sector has area 50 and arc length 10. Radius?

A. 20

B. 5

C. 2.5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

2319

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10

GRADING NOTE A = ½rs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

2320

QUESTION ID

S63-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A = ½r²θ and s = rθ, so A = ½r(rθ) = ½rs.

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

2322

QUESTION ID
S63-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4π − 8

GRADING NOTE Form required: exact value. Sector 4π minus right triangle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

2323

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
85 m²

GRADING NOTE Rounding to the nearest whole number required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

True or false?

The area of a circular segment (smaller than a semicircle) equals the sector area plus the area of the triangle formed by the two radii and the chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

2324

QUESTION ID
S63-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Subtract the triangle: segment = sector − triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

A. 8π

B. 4π

C. 16π

D. 25π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

2325

QUESTION ID

S63-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 16π

GRADING NOTE 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

2326

QUESTION ID

S63-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3351 cm²

GRADING NOTE Rounding to the nearest whole number required. $\frac{1}{3}\pi(60^2 - 20^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

Derive the equation of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Distance between two points with the Pythagorean Theorem	**8.G.B.8**
BACKWARD 2	Use the Pythagorean Theorem with expressions	**8.G.B.7**
BACKWARD 3	Shifting a graph: reading h and k	**HSF-BF.B.3**
FORWARD 1	Write circle equations from other given information	**HSG-GPE.A.1**
FORWARD 2	Prove or disprove that points lie on a circle	**HSG-GPE.B.4**

SET 64 • HSG-GPE.A.1

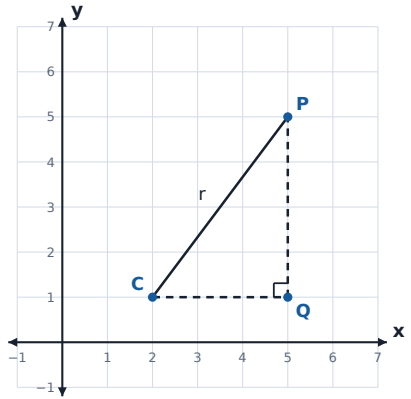
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (h, k) and radius r. Use the Pythagorean Theorem to derive the equation satisfied by every point (x, y) on the circle. The diagram shows one example, with center C(2, 1).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

2329

QUESTION ID

S64-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(x - h)^2 + (y - k)^2 = r^2$

GRADING NOTE REQUIRED WORK: Draw the right triangle with legs |x − h| and |y − k| and hypotenuse r.

Any valid proof (paragraph, two-column, or flow). Essential: a right triangle with horizontal leg |x − h|, vertical leg |y − k|, hypotenuse r (the distance CP); the Pythagorean Theorem gives $(x - h)^2 + (y - k)^2 = r^2$; squaring removes the absolute values; the reasoning holds for every point on the circle (degenerate legs of length 0 included). Invalid: only checking the one example point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (3, −2) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

2330

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2 + (y + 2)^2 = 25$

GRADING NOTE Form required: an equation; equivalent expanded forms accepted. Sign check: $y - (-2) = y + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center at the origin that passes through (6, −8).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

2331

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 100$

GRADING NOTE Form required: an equation. $r^2 = 36 + 64 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

Write the equation of the circle with center $(-1, 4)$ that passes through $(2, 8)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

2332

QUESTION ID

S64-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2 + (y - 4)^2 = 25$

GRADING NOTE Form required: an equation. $r^2 = 3^2 + 4^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the equation of the circle with center (−2, 5) and radius 3?

A. $(x - 2)^2 + (y + 5)^2 = 9$

B. $(x + 2)^2 + (y - 5)^2 = 9$

C. $(x + 2)^2 + (y - 5)^2 = 3$

D. $(x - 2)^2 + (y - 5)^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

2333

QUESTION ID

S64-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $(x + 2)^2 + (y - 5)^2 = 9$

GRADING NOTE Signs are opposite to the center coordinates; the right side is r^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 1

Distance between two points with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

2335

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

2336

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

The legs of the right triangle joining (2, 1) and (7, 13) are

A. 2 and 1

B. 7 and 13

C. 9 and 14

D. 5 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

2337

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5 and 12

GRADING NOTE Differences of coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

True or false?

The distance from $(-3, 0)$ to $(3, 0)$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

2338

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The distance from (0, 0) to (3, 4) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

2339

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5; the legs are not added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 2

Use the Pythagorean Theorem with expressions • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 6 and 8. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

2341

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs a and b and hypotenuse c.
Write the Pythagorean relationship.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

2342

QUESTION ID
S64-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$a^2 + b^2 = c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

A right triangle has legs $(x - 2)$ and $(y - 1)$ and hypotenuse 4. Which equation is true?

A. $(x - 2)^2 + (y - 1)^2 = 16$

B. $x^2 + y^2 = 16$

C. $(x - 2)^2 + (y - 1)^2 = 4$

D. $(x - 2) + (y - 1) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

2343

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 2)^2 + (y - 1)^2 = 16$

GRADING NOTE Square the legs and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

True or false?

In a right triangle, the hypotenuse is opposite the right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

2344

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ holds for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

2345

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 3

Shifting a graph: reading h and k • High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3

The graph of $y = (x - 3)^2$ is the graph of $y = x^2$ shifted in which direction and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

2347

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Right 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write x − (−4) without the double sign.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

2348

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x + 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

In $(x + 2)^2$, the shift is

A. 2 down

B. 2 right

C. 2 left

D. 2 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

2349

QUESTION ID

S64-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 left

GRADING NOTE $x - (-2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

True or false?

Replacing x with x − 5 shifts a graph 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

2350

QUESTION ID

S64-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Replacing y with y + 1 shifts a graph 1 unit up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

2351

QUESTION ID

S64-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shifts it 1 unit down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 1

Write circle equations from other given information • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A diameter of a circle has endpoints (−2, 3) and (6, −3). Write the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

2353

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 + y^2 = 25$

GRADING NOTE Form required: an equation. Center (2, 0) is the midpoint; radius 5 (half of 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (4, −3) and is tangent to the x-axis. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

2354

QUESTION ID
S64-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 4)^2 + (y + 3)^2 = 9$

GRADING NOTE Form required: an equation. Radius = distance to the x-axis = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 2$ passes through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

2355

QUESTION ID

S64-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

A circle with center (0, 5) passes through (0, −1). Its radius is

A. 4

B. 36

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

2356

QUESTION ID

S64-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Vertical distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at (3, 3) and is tangent to both axes. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

2357

QUESTION ID
S64-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y - 3)^2 = 9$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 2

Prove or disprove that points lie on a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Does (−4, 6) lie on the circle with center (2, −2) passing through (8, 6)? Show the check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

2359

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: both points are 10 from the center (36 + 64 = 100).

GRADING NOTE Form required: the conclusion with the distance check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (7, 1) inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

2360

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

On (16 + 9 = 25)

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

True or false?

A point is inside a circle when its distance from the center is greater than the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

2361

QUESTION ID

S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Greater than the radius means outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (1, 1) is ___ the circle $x^2 + y^2 = 4$.

A. on

B. outside

C. inside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

2362

QUESTION ID

S64-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. inside

GRADING NOTE $1 + 1 = 2 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

A circle has the diameter with endpoints (−2, 3) and (6, −3). Does (5, 4) lie on it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

2363

QUESTION ID

S64-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes:

$(5 - 2)^2 + 4^2 = 25$.

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

Complete the square to find the center and radius of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Complete the square in one variable	**HSA-REI.B.4.a**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
BACKWARD 3	Keep an equation balanced	**HSA-REI.A.1**
FORWARD 1	Leading coefficients and equations that are not circles	**HSG-GPE.A.1**
FORWARD 2	Complete the square for an ellipse	**HSG-GPE.A.3**

SET 65 • HSG-GPE.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

2366

QUESTION ID
S65-M-Q1

TYPE
Short answer

CORRECT ANSWER
Center (3, −2); radius 5

GRADING NOTE Form required: both. $(x - 3)^2 + (y + 2)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

Find the center and radius of the circle
$x^2 + y^2 + 8x - 10y + 5 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

2367

QUESTION ID
S65-M-Q2

TYPE
Short answer

CORRECT ANSWER
Center (−4, 5); radius 6

GRADING NOTE Form required: both. $(x + 4)^2 + (y - 5)^2 = -5 + 16 + 25 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 2x - 14y + 25 = 0$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

2368

QUESTION ID

S65-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(x - 1)^2 + (y - 7)^2 = 25$; center (1, 7); radius 5

GRADING NOTE REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 1 and 49 to both sides: $-25 + 1 + 49 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

Find the center and radius of the circle
$x^2 + y^2 + 10y = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

2369

QUESTION ID
S65-M-Q4

TYPE
Short answer

CORRECT ANSWER
Center (0, −5); radius 5

GRADING NOTE Form required: both. $x^2 + (y + 5)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

What is the radius of the circle $x^2 + y^2 - 4x + 6y = 3$?

A. 2

B. $\sqrt{3}$

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

2370

QUESTION ID

S65-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $(x - 2)^2 + (y + 3)^2 = 3 + 4 + 9 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 1

Complete the square in one variable • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

What number completes the square: $x^2 - 6x + ___?$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

2372

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $(6/2)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $x^2 + 10x + 25$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

2373

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 5)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 8x + 16$ equals

A. $(x + 4)^2$

B. $(x - 8)^2$

C. $(x - 4)^2$

D. $(x - 16)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

2374

QUESTION ID
S65-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $(x - 4)^2$

GRADING NOTE Half of 8 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To complete $x^2 + bx$, add $(b/2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

2375

QUESTION ID
S65-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 6x + 6$ is a perfect square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

2376

QUESTION ID
S65-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It needs + 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 2

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

2378

QUESTION ID
S65-B2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

2379

QUESTION ID
S65-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$y^2 + 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 5)^2 =$

A. $x^2 + 10x + 25$

B. $x^2 + 25$

C. $x^2 + 10x + 10$

D. $x^2 + 5x + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

2380

QUESTION ID
S65-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $x^2 + 10x + 25$

GRADING NOTE Middle term 2 · 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

2381

QUESTION ID
S65-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

2382

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing the middle term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 3

Keep an equation balanced • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If 9 is added to the left side of an equation, what must be done to the right side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

2384

QUESTION ID
S65-B3-Q1

TYPE
Short answer

CORRECT ANSWER
Add 9

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 6x = 7$. Add 9 to both sides and factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

2385

QUESTION ID

S65-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2 = 16$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 + y^2 - 4x = 5$ becomes $(x - 2)^2 + y^2 =$

A. 4

B. 5

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

2386

QUESTION ID

S65-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE 5 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

True or false?

Adding the same number to both sides keeps the solutions the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

2387

QUESTION ID

S65-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equality property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

True or false?

A number added to one side need not be added to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

2388

QUESTION ID
S65-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That changes the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 1

Leading coefficients and equations that are not circles • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the center and radius of
$2x^2 + 2y^2 - 8x + 12y - 6 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

2390

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (2, −3); radius 4

GRADING NOTE Form required: both. Divide by 2 first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $x^2 + y^2 + 2x + 4y + 10 = 0$ represent a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

2391

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it becomes $(x + 1)^2 + (y + 2)^2 = -5$, and a sum of squares cannot be negative.

GRADING NOTE Form required: the conclusion with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

True or false?

$x^2 + y^2 - 6x + 9 = 0$ represents a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

2392

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 3)^2 + y^2 = 0$: only (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3x^2 + 3y^2 - 12y = 0$ has radius

A. $\sqrt{12}$

B. 2

C. 12

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

2393

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $x^2 + (y - 2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For what values of c is $x^2 + y^2 - 4x + c = 0$ a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

2394

QUESTION ID

S65-F1-Q5

TYPE

Short answer

CORRECT ANSWER

c < 4

GRADING NOTE Form required: an inequality. $r^2 = 4 - c > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

2394

SET 65 • HSG-GPE.A.1

FORWARD 2

Complete the square for an ellipse • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $4x^2 + 9y^2 - 16x + 18y - 11 = 0$ in standard form and give the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

2396

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2/9 + (y + 1)^2/4 = 1$; center $(2, -1)$

GRADING NOTE Form required: both. $4(x - 2)^2 + 9(y + 1)^2 = 11 + 16 + 9 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the center of $x^2 + 4y^2 + 6x - 8y + 9 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

2397

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(−3, 1)

GRADING NOTE Form required: an ordered pair. $(x + 3)^2 + 4(y - 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

True or false?

After factoring 4 out of $4x^2 - 16x$, add 16 inside the parentheses to complete the square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

2398

QUESTION ID
S65-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Add 4 inside: $4(x^2 - 4x + 4)$, which adds 16 to the expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is NOT a circle?

A. $x^2 + 4y^2 = 4$

B. $2x^2 + 2y^2 = 8$

C. $x^2 + y^2 = 4$

D. $(x - 1)^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

2399

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 4y^2 = 4$

GRADING NOTE Unequal coefficients give an ellipse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $x^2 + 4y^2 = 4$ is not a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

2400

QUESTION ID

S65-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The x^2 and y^2 coefficients differ, so the "radius" differs by direction (2 across, 1 up).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5